

CBMR: Expanded Mathematical Derivations

Likelihoods, Scores, Hessians, Structured Covariance and Inference for Constant and Spatially Varying Effects

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Abstract

This is a standalone working appendix for manual mathematical review. It derives the Poisson and negative-binomial likelihood derivatives from their probability mass functions, then projects them into regression coefficients for both globally constant and spatially varying covariate effects. Independent-cell, moment-matched aggregated, and study-clustered negative-binomial models are kept distinct. Subsequent sections derive structured covariance, nuisance adjustment, penalties, robust covariance, and contrasts by explicit matrix algebra. Exact identities are distinguished from distributional approximations, numerical rearrangements, and asymptotic inference. The source is handwritten exposition: existing repository proofs support selected identities, not certification of every line of this document.

Reading convention. Throughout, a “Hessian of the log-likelihood” means $\nabla^2 \ell$, whereas $H = -\nabla^2 \ell$ is the *observed information*. Most final block formulas are written for H ; negate them to obtain the log-likelihood Hessian. Dispersion is fixed in regression derivatives unless explicitly stated otherwise.

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1 Scope, assumptions, and a map of the models

1.1 What is expanded, and what is not being asserted

The reference inputs are the supplied `cbmr_paper.pdf`, the published CBMR article `imag.a.1057.pdf` (doi:[10.1162/IMAG.a.1057](https://doi.org/10.1162/IMAG.a.1057)), and the existing Hessian, covariance, and integrated appendix notes in this repository. This document is separate from those PDFs. It develops the calculations rather than reproducing the articles.

Four qualifications apply throughout. First, differentiation of a chosen likelihood can be exact even if the likelihood is an approximation to the scientific data-generating process. Second, an information inverse is an asymptotic covariance approximation, not generally the exact finite-sample variance of an estimator. Third, a faster factorization changes neither identifiability nor the validity of the model. Fourth, the mathematical scope is broader than the currently exposed implementation API; an equation below does not imply that a corresponding model has been implemented.

1.2 Indices and dimensions

Symbol	Dimension	Definition
i, j, g	indices	Study, voxel, group
M, N, G	scalars	Numbers of studies, voxels, groups
$\mathcal{I}_g$	set	Studies in group g
Y_{ij}, μ_{ij}	scalars	Count and its positive mean
o_{ij}	scalar	Known, parameter-independent log offset
B, B_j	$N \times P, P \times 1$	Fixed basis matrix and transpose of row j
z_i, γ	$Q \times 1$	Global covariates and global coefficients
Z_i	$R \times 1$	Design row for spatial coefficient maps, as a column
ξ_g	$P \times 1$	Group baseline spatial coefficients
β_r, β	$P \times 1, RP \times 1$	One spatial coefficient map and stacked maps
θ	$n_\theta \times 1$	All regression coefficients for the chosen design
x_{ij}, X_i	$n_\theta \times 1, N \times n_\theta$	Cell design column and study design matrix
α, r, ν	scalars	Dispersion and inverse-dispersion shapes
H, I	$n_\theta \times n_\theta$	Observed and expected information

Vectors are columns; row j of X_i is $x_{ij}^\top$. The scalar n_θ denotes the total regression dimension in the current model; the covariance chapters use $d = n_\theta - Q$ for its spatial part. Q counts global effects, whereas R counts spatial maps; they are not interchangeable.

1.3 Globally constant versus spatially varying effects

For GC-CBMR, write

$$\eta_{ij} = \log \mu_{ij} = o_{ij} + B_j^\top \xi_{g(i)} + z_i^\top \gamma, \quad \theta = (\xi_1^\top, \dots, \xi_G^\top, \gamma^\top)^\top. \quad (1.1)$$

Here $n_\theta = GP + Q$. Let e_g be column g of the G -dimensional identity. Then

$$x_{ij} = \begin{pmatrix} e_{g(i)} \otimes B_j \\ z_i \end{pmatrix}. \quad (1.2)$$

The effect of global covariate z_{iq} on η_{ij} is γ_q , independent of j , even though the baseline $B_j^\top \xi_g$ varies spatially. “Globally constant” describes the *effect*, not the observed covariate.

For SV-CBMR, stack $\beta = (\beta_1^\top, \dots, \beta_R^\top)^\top$ and use

$$\eta_{ij} = o_{ij} + \sum_{r=1}^R Z_{ir} B_j^\top \beta_r, \quad x_{ij} = Z_i \otimes B_j, \quad (1.3)$$

$$b_r(j) = B_j^\top \beta_r, \quad \frac{\partial \eta_{ij}}{\partial Z_{ir}} = b_r(j). \quad (1.4)$$

Group indicator columns can be included in Z to supply baseline maps. Study index varies slowest in the stacked cell vector, giving $X = Z \otimes B$. A hybrid model simply appends $z_i^\top \gamma$ to (1.3), with design $(Z_i \otimes B_j, z_i)$. All affine chain rules below cover that hybrid without further assumptions.

For one moderator a_i , a matched comparison is

$$\eta_{ij}^{\text{GC}} = B_j^\top \xi_{g(i)} + a_i \gamma_a, \quad \eta_{ij}^{\text{SV}} = B_j^\top \xi_{g(i)} + a_i B_j^\top \beta_a.$$

One unit of a_i multiplies the mean by e^{γ_a} in the first case and by $e^{B_j^\top \beta_a}$ in the second. If $Bc_B = \mathbf{1}$, the restriction $\beta_a = c_B \gamma_a$ embeds GC in SV. It is constancy of $B\beta_a$, not necessarily equality of the entries of β_a , that matters.

1.4 The observation model is a separate choice

Family	Stochastic specification	Scope of reduction
Poisson	Independent Y_{ij} with variance μ_{ij}	Affine GC/SV Hessian; exact grouped compression when means separate
Independent NB	Independent Y_{ij} with variance $\mu_{ij} + \alpha_g \mu_{ij}^2$	Affine GC/SV Hessian; generally requires cell-level information
Aggregated NB	NB working distribution fitted to group-voxel totals	Moment matching is usually approximate; grouped separability simplifies derivatives
Clustered NB	One gamma random multiplier shared by all voxels of a study	Exact integration; affine GC/SV Hessian has a rank-one correction

The classes named `NegativeBinomial` in the supplied `cbmr_code` and `cbmr_improved` use the *aggregated* model, not the independent-cell likelihood. The present implementation requires shared spatial patterns for its overdispersed distributions; its restriction is not a mathematical prohibition on an independent-cell or clustered-NB SV model.

We assume a fixed, known design and offsets, positive finite means, and nonnegative integer counts. With binary observations, a Poisson or NB likelihood is a working count model, not a Bernoulli likelihood. Identifiability and nonsingular information are additional requirements when taking inverses. Merely specifying an affine predictor does not ensure a finite maximum-likelihood estimate.

2 Chain rules: from scalar likelihood to coefficient Hessian

2.1 The elementary log-link calculation

For a scalar cell write $\eta = o + x^\top \theta$ and $\mu = e^\eta$. For coefficient indices a, b ,

$$\frac{\partial \eta}{\partial \theta_a} = x_a, \quad \frac{\partial^2 \eta}{\partial \theta_a \partial \theta_b} = 0, \quad (2.1)$$

$$\frac{\partial \mu}{\partial \theta_a} = \mu x_a, \quad \frac{\partial^2 \mu}{\partial \theta_a \partial \theta_b} = \mu x_a x_b. \quad (2.2)$$

If a log-likelihood summand is initially written as $f(\mu)$,

$$\frac{\partial f}{\partial \theta_a} = f_{\mu} \mu x_a, \quad (2.3)$$

$$\frac{\partial^2 f}{\partial \theta_a \partial \theta_b} = f_{\mu\mu} \mu^2 x_a x_b + f_{\mu} \mu x_a x_b. \quad (2.4)$$

Equivalently $f_{\eta} = \mu f_{\mu}$ and $f_{\eta\eta} = \mu^2 f_{\mu\mu} + \mu f_{\mu}$. The second term in (2.4) must not be lost by treating the exponential mean as affine.

2.2 Independent-cell assembly, entry by entry

Let $\ell(\theta) = \sum_{i,j} f_{ij}(\eta_{ij})$ and set $s_{ij} = f'_{ij}(\eta_{ij})$ and $w_{ij} = -f''_{ij}(\eta_{ij})$. Taking one derivative and then a second gives

$$U_a = \frac{\partial \ell}{\partial \theta_a} = \sum_{i,j} s_{ij} x_{ij,a}, \quad (2.5)$$

$$\frac{\partial U_a}{\partial \theta_b} = \sum_{i,j} \left[f''_{ij} x_{ij,b} x_{ij,a} + s_{ij} \frac{\partial x_{ij,a}}{\partial \theta_b} \right] = - \sum_{i,j} w_{ij} x_{ij,a} x_{ij,b}. \quad (2.6)$$

The last equality uses the *fixed affine* design. Thus

$$U = \sum_i X_i^{\top} s_i, \quad H = \sum_i X_i^{\top} \text{diag}(w_i) X_i, \quad \nabla^2 \ell = -H. \quad (2.7)$$

For a nonlinear predictor there would instead be the extra term $-\sum_{i,j} s_{ij} \nabla_{\theta}^2 \eta_{ij}$ in H .

2.3 GC and SV block entries

For GC, one study contributes only to its own group baseline and to the global coefficients. Consequently

$$H_{\xi_g \xi_g} = \sum_{i \in \mathcal{I}_g} \sum_j w_{ij} B_j B_j^{\top}, \quad (2.8)$$

$$H_{\xi_g \gamma} = \sum_{i \in \mathcal{I}_g} \sum_j w_{ij} B_j z_i^{\top}, \quad (2.9)$$

$$H_{\gamma \gamma} = \sum_g \sum_{i \in \mathcal{I}_g} \sum_j w_{ij} z_i z_i^{\top}, \quad H_{\xi_g \xi_h} = 0 \quad (g \neq h). \quad (2.10)$$

Each displayed cross block has size $P \times Q$; its transpose is $H_{\gamma \xi_g}$. Zero spatial cross blocks do not imply zero cross-group *covariance*.

For SV, $x_{ij,(r,p)} = Z_{ir} B_{jp}$, so

$$H_{(r,p),(r',p')} = \sum_i Z_{ir} Z_{ir'} \sum_j w_{ij} B_{jp} B_{jp'}, \quad (2.11)$$

$$H_{\beta_r \beta_{r'}} = \sum_i Z_{ir} Z_{ir'} B^{\top} \text{diag}(w_i) B, \quad (2.12)$$

$$H = \sum_i (Z_i Z_i^{\top}) \otimes \{B^{\top} \text{diag}(w_i) B\}. \quad (2.13)$$

This is a *sum* of Kronecker products, not generally a single Kronecker product. It is valid for any weights obtained from an independent-cell likelihood, without separability of the means.

2.4 Joint study likelihoods

If ℓ_i depends jointly on $\eta_i = (\eta_{i1}, \dots, \eta_{iN})^\top$, define $s_i = \nabla_{\eta_i} \ell_i$ and $W_i = -\nabla_{\eta_i}^2 \ell_i$. Since $\eta_i = o_i + X_i \theta$,

$$U = \sum_i X_i^\top s_i, \quad H = \sum_i X_i^\top W_i X_i. \quad (2.14)$$

Here W_i need not be diagonal. This distinction will be essential for clustered NB, whose cells are dependent after integrating its random effect.

3 Poisson: full scalar derivation and GC/SV Hessians

3.1 Mass function, score, and second derivative

For $y = 0, 1, 2, \dots$,

$$p(y \mid \mu) = e^{-\mu} \frac{\mu^y}{y!}, \quad \ell = y\eta - e^\eta - \log \Gamma(y+1).$$

Differentiate each term:

$$\ell_\eta = y - e^\eta = y - \mu, \quad (3.1)$$

$$\ell_{\eta\eta} = -e^\eta = -\mu, \quad (3.2)$$

$$U = \sum_{i,j} x_{ij} (Y_{ij} - \mu_{ij}), \quad (3.3)$$

$$H = \sum_{i,j} \mu_{ij} x_{ij} x_{ij}^\top. \quad (3.4)$$

The counts disappear on the second derivative at a *fixed* parameter value. $H(\widehat{\theta})$ still depends on the data through $\widehat{\theta}$. Since H already has no explicit random Y , $I = E_\theta H = H$. Independence also yields

$$\text{Var}_\theta(U) = \sum_{i,j} x_{ij} x_{ij}^\top \text{Var}(Y_{ij}) = \sum_{i,j} \mu_{ij} x_{ij} x_{ij}^\top = I.$$

This last equality is statistical and uses the Poisson independence assumption; differentiating the likelihood did not require the real data to obey that assumption.

3.2 Exact grouped compression for GC

To expose separability, suppose $o_{ij} = \log e_i$ with known $e_i > 0$; set

$$u_{gj} = e^{B_j^\top \xi_g}, \quad v_i = e_i e^{z_i^\top \gamma}, \quad \mu_{ij} = u_{g(i)j} v_i.$$

A fixed separable voxel offset can also be absorbed into u_{gj} . Arbitrary nonseparable o_{ij} is allowed in the full-cell formulas, but not in the following reduced formulas. Define the two count marginals and mean totals

$$Y_{gj} = \sum_{i \in \mathcal{I}_g} Y_{ij}, \quad n_i = \sum_j Y_{ij}, \quad A_g^{(u)} = \sum_j u_{gj}, \quad T_g = \sum_{i \in \mathcal{I}_g} v_i.$$

The symbol $A_g^{(u)}$ here is just the spatial intensity total; it is different from the NB parameter A introduced later. Substitute the mean into the *full-cell* likelihood:

$$\ell = \sum_{i,j} Y_{ij} \{ \log e_i + B_j^\top \xi_{g(i)} + z_i^\top \gamma \} - \sum_g \sum_{i \in \mathcal{I}_g} \sum_j v_i u_{gj} + c(Y) \quad (3.5)$$

$$= \sum_g \sum_j Y_{gj} B_j^\top \xi_g + \sum_i n_i z_i^\top \gamma - \sum_g A_g^{(u)} T_g + c(Y, e). \quad (3.6)$$

Both count marginals are retained. The $n_i \log e_i$ terms may be dropped because exposure is fixed; the $n_i z_i^\top \gamma$ terms may not.

Write

$$a_g = B^\top u_g, \quad K_g = B^\top \text{diag}(u_g) B, \quad t_g = \sum_{i \in \mathcal{I}_g} v_i z_i, \quad M_g^{(v)} = \sum_{i \in \mathcal{I}_g} v_i z_i z_i^\top.$$

Their differential identities follow directly from the exponential:

$$\nabla_{\xi_g} A_g^{(u)} = a_g, \quad \nabla_{\xi_g}^2 A_g^{(u)} = K_g, \quad \nabla_\gamma T_g = t_g, \quad \nabla_\gamma^2 T_g = M_g^{(v)}.$$

Differentiate (3.6) once:

$$U_{\xi_g} = B^\top Y_g - T_g a_g, \tag{3.7}$$

$$U_\gamma = \sum_i n_i z_i - \sum_g A_g^{(u)} t_g. \tag{3.8}$$

Differentiating those scores gives, including their signs,

$$\frac{\partial U_{\xi_g}}{\partial \xi_g^\top} = -T_g K_g, \quad \frac{\partial U_{\xi_g}}{\partial \gamma^\top} = -a_g t_g^\top, \tag{3.9}$$

$$\frac{\partial U_\gamma}{\partial \gamma^\top} = -\sum_g A_g^{(u)} M_g^{(v)}. \tag{3.10}$$

Negating gives

$$H_{\xi_g \xi_g} = T_g K_g, \quad H_{\xi_g \gamma} = a_g t_g^\top, \quad H_{\gamma \gamma} = \sum_g A_g^{(u)} M_g^{(v)}. \tag{3.11}$$

This reproduces (2.10) with $w_{ij} = v_i u_{gj}$, but never requires an $M \times N$ mean array.

3.3 Why fitting voxel totals alone is different

An aggregated Poisson fit instead assigns $Y_{gj} \sim \text{Poisson}(T_g u_{gj})$ and has

$$\ell_{\text{totals}} = \sum_{g,j} Y_{gj} \{\log T_g + \log u_{gj}\} - \sum_g T_g A_g^{(u)} + c(Y).$$

Subtracting from the full likelihood gives, up to data-only terms,

$$\ell_{\text{full}} - \ell_{\text{totals}} = \sum_g \sum_{i \in \mathcal{I}_g} n_i \log(v_i / T_g). \tag{3.12}$$

These are the conditional multinomial allocation terms across studies. For a group with $n_g = \sum_i n_i$, their global score and negative Hessian are

$$U_{\text{alloc},g} = \sum_{i \in \mathcal{I}_g} n_i z_i - n_g t_g / T_g, \tag{3.13}$$

$$H_{\text{alloc},g} = n_g \left\{ M_g^{(v)} / T_g - t_g t_g^\top / T_g^2 \right\}. \tag{3.14}$$

The bracket is the covariance of z_i under probabilities v_i / T_g , so it is positive semidefinite. The aggregated-NB Poisson limit recovers the totals likelihood, not automatically the full likelihood with these extra terms.

3.4 SV assembly and exact Kronecker factorization when available

For SV substitute $w_{ij} = \mu_{ij}$ into (2.13):

$$H_{\beta_r, \beta_{r'}} = \sum_i Z_{ir} Z_{ir'}^\top B^\top \text{diag}(\mu_i) B.$$

If, at the parameter value in question, $\mu_{ij} = a_i b_j$, then

$$H = \sum_i a_i Z_i Z_i^\top \otimes B^\top \text{diag}(b) B \quad (3.15)$$

$$= \{Z^\top \text{diag}(a) Z\} \otimes \{B^\top \text{diag}(b) B\}. \quad (3.16)$$

General SV effects do not make μ separable. Equation (3.16) is the information for the RP -dimensional SV design *evaluated at a separable mean*; it is not the $(GP + Q)$ -dimensional GC information (3.11). A factorization computed at an initialization is a preconditioner if later means are nonseparable, not the exact later Hessian.

4 Independent-cell negative binomial: every scalar derivative

4.1 Parameterization and probability model

Let $\alpha > 0$ and $r = \alpha^{-1}$. Use the NB2 convention

$$p(y \mid \mu, r) = \frac{\Gamma(y+r)}{\Gamma(r)\Gamma(y+1)} \left(\frac{r}{r+\mu}\right)^r \left(\frac{\mu}{r+\mu}\right)^y, \quad \mathbb{E} Y = \mu, \quad \text{Var } Y = \mu + \mu^2/r. \quad (4.1)$$

One construction is $\Lambda \sim \text{Gamma}(r, \text{rate } r)$, $Y \mid \Lambda \sim \text{Poisson}(\Lambda\mu)$. Independent NB cells have independent such multipliers. The log-likelihood is

$$\ell = \log \Gamma(y+r) - \log \Gamma(r) - \log \Gamma(y+1) + r \log r + y\eta - (y+r) \log(r + e^\eta) \quad (4.2)$$

$$= \log \Gamma(y + \alpha^{-1}) - \log \Gamma(\alpha^{-1}) - \log \Gamma(y+1) + y \log \alpha + y\eta - (y + \alpha^{-1}) \log(1 + \alpha e^\eta). \quad (4.3)$$

All displayed terms must be retained if dispersion is differentiated. The gamma terms are constant only for the fixed-dispersion regression step.

4.2 Score: differentiate, combine the numerator, then simplify

Because $\partial_\eta \log(r + e^\eta) = \mu/(r + \mu)$,

$$\ell_\eta = y - (y+r) \frac{\mu}{r+\mu} \quad (4.4)$$

$$= \frac{y(r+\mu) - (y+r)\mu}{r+\mu} = \frac{r(y-\mu)}{r+\mu} = \frac{y-\mu}{1+\alpha\mu}. \quad (4.5)$$

Thus the NB score is a weighted residual, not the Poisson residual.

4.3 Second derivative: explicit quotient rule

Use $\partial_\eta(y - \mu) = -\mu$ and $\partial_\eta(1 + \alpha\mu) = \alpha\mu$:

$$\ell_{\eta\eta} = \frac{(-\mu)(1 + \alpha\mu) - (y - \mu)\alpha\mu}{(1 + \alpha\mu)^2} \quad (4.6)$$

$$= \frac{-\mu - \alpha\mu^2 - \alpha\mu y + \alpha\mu^2}{(1 + \alpha\mu)^2} \quad (4.7)$$

$$= -\frac{\mu(1 + \alpha y)}{(1 + \alpha\mu)^2} = -\frac{(r+y)r\mu}{(r+\mu)^2}. \quad (4.8)$$

Therefore

$$w^{\text{obs}} = \frac{\mu(1 + \alpha\mu)}{(1 + \alpha\mu)^2}, \quad w^{\text{F}} = \mathbb{E} w^{\text{obs}} = \frac{\mu(1 + \alpha\mu)}{(1 + \alpha\mu)^2} = \frac{\mu}{1 + \alpha\mu}. \quad (4.9)$$

The expectation holds μ, α fixed and uses $\mathbb{E} Y = \mu$. Independently,

$$\text{Var}(\ell_\eta) = \frac{\text{Var} Y}{(1 + \alpha\mu)^2} = \frac{\mu + \alpha\mu^2}{(1 + \alpha\mu)^2} = w^{\text{F}}.$$

At fixed dispersion both weights are strictly positive for finite positive μ . Hence $H = X^\top \text{diag}(w^{\text{obs}})X$ is positive semidefinite, and positive definite if the stacked design has full column rank. This is fixed-dispersion convexity of $-\ell$, not joint convexity in regression and dispersion.

4.4 GC and SV formulas, with no change of likelihood

For group-dependent dispersion set $\alpha = \alpha_{g(i)}$ in each cell. The score and observed information under either affine design are

$$U = \sum_{i,j} x_{ij} \frac{Y_{ij} - \mu_{ij}}{1 + \alpha_{g(i)}\mu_{ij}}, \quad (4.10)$$

$$H = \sum_{i,j} x_{ij} x_{ij}^\top \frac{\mu_{ij}(1 + \alpha_{g(i)}Y_{ij})}{(1 + \alpha_{g(i)}\mu_{ij})^2}, \quad (4.11)$$

$$I = \sum_{i,j} x_{ij} x_{ij}^\top \frac{\mu_{ij}}{1 + \alpha_{g(i)}\mu_{ij}}. \quad (4.12)$$

For GC insert these weights into (2.10). For SV, for example, the fully expanded coefficient entry is

$$H_{(r,p),(r',p')} = \sum_i \sum_j Z_{ir} Z_{ir'} B_{jp} B_{jp'} \frac{\mu_{ij}(1 + \alpha_{g(i)}Y_{ij})}{(1 + \alpha_{g(i)}\mu_{ij})^2}. \quad (4.13)$$

Unlike Poisson, the observed Hessian depends explicitly on the cell counts. Even for GC with $\mu_{ij} = u_{gj}v_i$, the denominators prevent the simple products in (3.11).

4.5 Why separable NB means usually do not give separable weights

Take a common $\alpha > 0$, study factors $a, b > 0$, and voxel factors $c, d > 0$. The determinant of their 2×2 Fisher weight grid is

$$\frac{ac}{1 + \alpha ac} \frac{bd}{1 + \alpha bd} - \frac{ad}{1 + \alpha ad} \frac{bc}{1 + \alpha bc} \quad (4.14)$$

$$= \frac{abcd\{(1 + \alpha ad)(1 + \alpha bc) - (1 + \alpha ac)(1 + \alpha bd)\}}{(1 + \alpha ac)(1 + \alpha ad)(1 + \alpha bc)(1 + \alpha bd)} \quad (4.15)$$

$$= \frac{\alpha abcd(a - b)(d - c)}{(1 + \alpha ac)(1 + \alpha ad)(1 + \alpha bc)(1 + \alpha bd)}. \quad (4.16)$$

This is nonzero when $a \neq b$ and $c \neq d$. A separable grid has rank one and every such determinant zero. Thus separability of μ alone does not justify the Poisson Kronecker shortcut for NB information.

4.6 Fisher scoring, IRLS, and the Poisson limit

Linearizing the score equation near θ gives the Newton update $\theta_{\text{new}} = \theta + H^{-1}U$. Fisher scoring replaces H by I . For NB the working response is $z_{ij}^* = \eta_{ij} + (Y_{ij} - \mu_{ij})/\mu_{ij}$. Since the offset is fixed,

$$X^\top W^F(z^* - o - X\theta) = X^\top W^F(z^* - \eta) = \sum_{i,j} x_{ij} \frac{\mu_{ij}}{1 + \alpha\mu_{ij}} \frac{Y_{ij} - \mu_{ij}}{\mu_{ij}} = U.$$

This derives the equivalence of weighted least squares and the correct NB Fisher-scoring direction. Using unweighted Poisson residuals with NB Fisher weights is a different estimating equation.

Holding counts and means fixed as $\alpha \downarrow 0$,

$$s_{\text{NB}} \longrightarrow y - \mu, \quad w_{\text{NB}}^{\text{obs}} \longrightarrow \mu, \quad w_{\text{NB}}^F \longrightarrow \mu.$$

For $y = 0$ or 1 the gamma terms and $y \log \alpha$ in (4.3) cancel, but the result is still the NB mass assigned to a count, not a normalized Bernoulli model.

4.7 Dispersion derivatives when it is not held fixed

Let $\psi(t) = d \log \Gamma(t)/dt$ and $\psi_1(t) = d\psi(t)/dt$. Differentiating the first form of (4.3) at fixed μ :

$$\ell_r = \psi(y+r) - \psi(r) + \log r + 1 - \log(r+\mu) - \frac{y+r}{r+\mu}, \quad (4.17)$$

$$\ell_{rr} = \psi_1(y+r) - \psi_1(r) + \frac{1}{r} - \frac{1}{r+\mu} - \frac{\mu-y}{(r+\mu)^2}, \quad (4.18)$$

$$\ell_{\eta r} = \frac{\partial}{\partial r} \left\{ \frac{r(y-\mu)}{r+\mu} \right\} = \frac{\mu(y-\mu)}{(r+\mu)^2}. \quad (4.19)$$

For one shared shape, $H_{\theta r} = -\sum_{i,j} x_{ij} \ell_{\eta r}$ and $H_{rr} = -\sum_{i,j} \ell_{rr}$. Group-specific shapes restrict those sums to their groups. Under the correctly specified independent NB model, $E \ell_{\eta r} = 0$: regression and shape are orthogonal in expected information. Observed cross-information generally is not zero. Orthogonality does not justify ignoring nuisance terms in an arbitrary robust sandwich.

On the log-dispersion scale $\delta = \log \alpha$ we have $r = e^{-\delta}$. Applying the chain rule once and twice gives

$$\ell_\delta = -r\ell_r, \quad \ell_{\delta\delta} = r^2\ell_{rr} + r\ell_r, \quad \ell_{\theta\delta} = -r\ell_{\theta r}.$$

The $r\ell_r$ term cannot be discarded away from a stationary point. The boundary $\alpha = 0$ is not an interior log-dispersion value, so ordinary joint Wald arguments need separate treatment at that boundary.

5 Aggregated NB: moment matching and the full composite Hessian

5.1 Deriving the approximation from moments

Work within one GC group and omit g temporarily. Let $u_j = e^{B_j^\top \xi}$, $v_i = e_i e^{z_i^\top \gamma}$, and $Y_j = \sum_i Y_{ij}$. Suppose the underlying independent cells have variance $u_j v_i + \alpha u_j^2 v_i^2$. Define

$$s_1 = \sum_i v_i, \quad s_2 = \sum_i v_i^2, \quad s_3 = \sum_i v_i^3.$$

By independence across studies,

$$m_j = E Y_j = u_j s_1, \quad (5.1)$$

$$\text{Var } Y_j = \sum_i (u_j v_i + \alpha u_j^2 v_i^2) = u_j s_1 + \alpha u_j^2 s_2. \quad (5.2)$$

An NB with mean m_j and shape ρ has variance $m_j + m_j^2/\rho$. Equate the extra variances:

$$\frac{u_j^2 s_1^2}{\rho} = \alpha u_j^2 s_2 \implies \rho = \frac{s_1^2}{\alpha s_2}.$$

Define $A = s_1/(\alpha s_2)$, so $\rho = s_1 A$ and

$$p_j = \frac{m_j}{m_j + \rho} = \frac{u_j}{u_j + A}, \quad \alpha_{\text{eff}} = \rho^{-1} = \alpha s_2 / s_1^2. \quad (5.3)$$

The common spatial factor cancels from the shape. The approximation does *not* say that a general sum of unequal-probability NB variables is exactly NB.

For completeness, an NB with mean μ and dispersion α has cumulant-generating function $K(t) = -\alpha^{-1} \log\{1 - \alpha\mu(e^t - 1)\}$. Put $q(t) = 1 - \alpha\mu(e^t - 1)$. Then

$$K'(t) = \mu e^t / q(t), \quad (5.4)$$

$$K''(t) = \mu e^t / q(t) + \alpha \mu^2 e^{2t} / q(t)^2, \quad (5.5)$$

$$K'''(0) = \mu + 3\alpha\mu^2 + 2\alpha^2\mu^3. \quad (5.6)$$

Adding true cell cumulants and subtracting the matched NB cumulant gives

$$\kappa_3^{\text{true}} - \kappa_3^{\text{matched}} = 2\alpha^2 u_j^3 \left(s_3 - \frac{s_2^2}{s_1} \right). \quad (5.7)$$

Cauchy-Schwarz applied to $v_i^{1/2}$ and $v_i^{3/2}$ gives $s_2^2 \leq s_1 s_3$, with equality for positive v_i only when all v_i are equal. In that equal-factor case the NB convolution is exact; generically it is not.

5.2 Rewrite the working negative log-likelihood

Let $y_j = Y_j$, $\eta_j = \log u_j$, and $d_j = u_j + A$. The matched NB log-likelihood is

$$\ell_{\text{agg}} = \sum_j \{ \log \Gamma(y_j + \rho) - \log \Gamma(\rho) - \log \Gamma(y_j + 1) + \rho \log(1 - p_j) + y_j \log p_j \}.$$

Since $\log(1 - p_j) = \log A - \log d_j$ and $\log p_j = \eta_j - \log d_j$, its negative, omitting data-only terms, is

$$\begin{aligned} \Psi(\eta, \rho, A) = & - \sum_j \log \Gamma(y_j + \rho) + N \log \Gamma(\rho) - N \rho \log A \\ & + \sum_j (\rho + y_j) \log(u_j + A) - \sum_j y_j \eta_j. \end{aligned} \quad (5.8)$$

We first differentiate treating (η, ρ, A) as independent coordinates. Only afterwards do we compose $\rho(\gamma), A(\gamma)$. This two-stage calculation prevents omitted chain-rule terms.

5.3 Spatial and mixed partial derivatives

At fixed ρ, A ,

$$\Psi_{\eta_j} = (\rho + y_j) \frac{u_j}{d_j} - y_j, \quad (5.9)$$

$$\frac{\partial}{\partial \eta_j} \left(\frac{u_j}{d_j} \right) = \frac{u_j d_j - u_j^2}{d_j^2} = \frac{u_j A}{d_j^2}, \quad (5.10)$$

$$\Psi_{\eta_j \eta_j} = \frac{(\rho + y_j) u_j A}{d_j^2} = w_j, \quad \Psi_{\eta_j \eta_k} = 0 \quad (j \neq k), \quad (5.11)$$

$$\Psi_{\eta_j \rho} = \frac{u_j}{d_j}, \quad \Psi_{\eta_j A} = -\frac{(\rho + y_j) u_j}{d_j^2}. \quad (5.12)$$

Because $\eta = B\xi$ is affine,

$$U_\xi = -B^\top \Psi_\eta, \quad H_{\xi\xi} = B^\top \text{diag}(w)B.$$

These are derivatives of the matched likelihood, not the independent-cell Hessian with cells silently replaced by totals.

5.4 Outer derivatives in shape and scale

Differentiate (5.8), retaining its signs:

$$\Psi_\rho = -\sum_j \psi(y_j + \rho) + N\psi(\rho) - N \log A + \sum_j \log d_j, \quad (5.13)$$

$$\Psi_A = -N\rho/A + \sum_j (\rho + y_j)/d_j, \quad (5.14)$$

$$\Psi_{\rho\rho} = -\sum_j \psi_1(y_j + \rho) + N\psi_1(\rho), \quad (5.15)$$

$$\Psi_{\rho A} = -N/A + \sum_j d_j^{-1}, \quad (5.16)$$

$$\Psi_{AA} = N\rho/A^2 - \sum_j (\rho + y_j)/d_j^2. \quad (5.17)$$

For example the last line differentiates both $-N\rho/A$ and every d_j^{-1} in Ψ_A . These two contributions can be large with opposite signs; the log-coordinate alternative below reduces that cancellation.

5.5 Inner derivatives of the aggregate parameter map

The exponential moderator satisfies $\nabla_\gamma v_i = v_i z_i$ and $\nabla_\gamma^2 v_i = v_i z_i z_i^\top$. Squaring it changes the factors:

$$\nabla(v_i^2) = 2v_i^2 z_i, \quad \nabla^2(v_i^2) = 4v_i^2 z_i z_i^\top.$$

Put

$$u_1 = \sum_i v_i z_i, \quad u_2 = \sum_i v_i^2 z_i, \quad M_1 = \sum_i v_i z_i z_i^\top, \quad M_2 = \sum_i v_i^2 z_i z_i^\top.$$

Then

$$\nabla s_1 = u_1, \quad \nabla^2 s_1 = M_1, \quad (5.18)$$

$$\nabla s_2 = 2u_2, \quad \nabla^2 s_2 = 4M_2. \quad (5.19)$$

Applying $\nabla^2 \log s = s^{-1} \nabla^2 s - s^{-2} (\nabla s)(\nabla s)^\top$ gives

$$l = \nabla \log s_1 = u_1/s_1, \quad L_1 = \nabla^2 \log s_1 = M_1/s_1 - u_1 u_1^\top / s_1^2, \quad (5.20)$$

$$h = \nabla \log s_2 = 2u_2/s_2, \quad L_2 = \nabla^2 \log s_2 = 4M_2/s_2 - 4u_2 u_2^\top / s_2^2. \quad (5.21)$$

Since original α is fixed,

$$a = \log A = \log s_1 - \log s_2 - \log \alpha, \quad a_\gamma = l - h, \quad a_{\gamma\gamma} = L_1 - L_2, \quad (5.22)$$

$$q = \log \rho = 2 \log s_1 - \log s_2 - \log \alpha, \quad q_\gamma = 2l - h, \quad q_{\gamma\gamma} = 2L_1 - L_2. \quad (5.23)$$

For any $A = e^a$, differentiating $\nabla A = A \nabla a$ gives

$$\nabla A = A(l - h), \quad (5.24)$$

$$\nabla^2 A = A\{(l - h)(l - h)^\top + L_1 - L_2\}, \quad (5.25)$$

$$\nabla \rho = \rho(2l - h), \quad (5.26)$$

$$\nabla^2 \rho = \rho\{(2l - h)(2l - h)^\top + 2L_1 - L_2\}. \quad (5.27)$$

This explicitly identifies every first and second inner-map derivative.

5.6 Compose the derivatives: no missing curvature terms

For global indices a, b write $\rho_a = \partial_{\gamma_a} \rho$ and similarly A_a, ρ_{ab}, A_{ab} . The first derivative is $\partial_{\gamma_a} \Psi = \Psi_{\rho} \rho_a + \Psi_A A_a$. Differentiating both products with respect to γ_b yields

$$H_{\gamma_a \gamma_b} = \Psi_{\rho\rho} \rho_a \rho_b + \Psi_{\rho A} (A_b \rho_a + \rho_b A_a) + \Psi_{AA} A_a A_b \quad (5.28)$$

$$+ \Psi_{\rho} \rho_{ab} + \Psi_A A_{ab}. \quad (5.29)$$

The last two terms are essential. Ψ_{ρ} and Ψ_A need not separately vanish when the composite regression score is zero. Deleting those terms gives a different Hessian even at many fitted parameter values.

Differentiating $\nabla_{\xi} \Psi = B^{\top} \Psi_{\eta}$ with respect to γ gives the cross block

$$H_{\xi\gamma} = \sum_j B_j \left\{ \frac{u_j}{d_j} (\nabla \rho)^{\top} - \frac{(\rho + y_j) u_j}{d_j^2} (\nabla A)^{\top} \right\}. \quad (5.30)$$

The global score is $U_{\gamma} = -\Psi_{\rho} \nabla \rho - \Psi_A \nabla A$. For multiple groups each spatial block occupies its own position, and the group global scores and global Hessians are summed. Fixed original α_g does not mean fixed $\alpha_{\text{eff},g} = \alpha_g s_{2g} / s_{1g}^2$. Holding effective dispersion fixed would define different derivatives.

5.7 Expected information under the matched working distribution

If the matched NB distribution itself is the assumed model, its coordinate scores have zero expectation. In particular $E \Psi_{\rho} = E \Psi_A = 0$, so the last two terms of (5.29) vanish *in expectation*, not generally at an observed optimum. With $m_j = \rho u_j / A$, the expected outer partials are

$$E w_j = \frac{(\rho + m_j) u_j A}{d_j^2} = \frac{\rho u_j}{d_j}, \quad (5.31)$$

$$E \Psi_{\rho\rho} = N \psi_1(\rho) - \sum_j E \{ \psi_1(Y_{\cdot j} + \rho) \}, \quad (5.32)$$

$$E \Psi_{\rho A} = -N/A + \sum_j d_j^{-1}, \quad (5.33)$$

$$E \Psi_{AA} = N \rho / A^2 - \sum_j (\rho + m_j) / d_j^2, \quad (5.34)$$

$$E \Psi_{\eta_j \rho} = u_j / d_j, \quad E \Psi_{\eta_j A} = -(\rho + m_j) u_j / d_j^2. \quad (5.35)$$

The remaining expectation, if required numerically, is the NB-weighted sum $\sum_{y=0}^{\infty} \psi_1(y + \rho) p_{\text{NB}}(y; m_j, \rho)$. Insert these expected partials and the same deterministic inner derivatives into (5.29) and (5.30). There is no general license to replace $E \psi_1(Y + \rho)$ by $\psi_1(E Y + \rho)$.

Under the *original* independent-cell model the sum need not be NB. Moment matching fixes its first two moments, not its expected digamma term. Consequently the matched shape score need not have zero expectation under that original model. Since ρ depends on the global coefficients, the working regression score need not be centered at the original regression parameter either. An exact Hessian or a robust covariance for the working estimator does not remove this possible change in the parameter being estimated.

5.8 Stable log-coordinate rearrangement, derived explicitly

Let $q = \log \rho$, $a = \log A$, $t_j = \eta_j - a$, $p_j = (1 + e^{-t_j})^{-1}$, and $\bar{p}_j = 1 - p_j$. Using $\log(u_j + A) = a + \log(1 + e^{t_j})$,

$$\Psi = - \sum_j \log \Gamma(y_j + \rho) + N \log \Gamma(\rho) + \sum_j (\rho + y_j) \log(1 + e^{t_j}) - \sum_j y_j t_j. \quad (5.36)$$

At fixed η, q , $\partial_a t_j = -1$. At fixed η, a , $\partial_q \rho = \rho$. Thus

$$\Psi_a = \sum_j \{y_j - (\rho + y_j)p_j\} = \sum_j (y_j \bar{p}_j - \rho p_j), \quad (5.37)$$

$$\Psi_{aa} = \sum_j (\rho + y_j)p_j \bar{p}_j = \sum_j w_j, \quad (5.38)$$

$$\Psi_q = \rho \left\{ \sum_j \log(1 + e^{t_j}) - \sum_j [\psi(y_j + \rho) - \psi(\rho)] \right\}, \quad (5.39)$$

$$\Psi_{qa} = -\rho \sum_j p_j, \quad (5.40)$$

$$\Psi_{qq} = \Psi_q + \rho^2 \sum_j \{\psi_1(\rho) - \psi_1(y_j + \rho)\}, \quad (5.41)$$

$$\Psi_{\eta_j q} = \rho p_j, \quad \Psi_{\eta_j a} = -(\rho + y_j)p_j \bar{p}_j = -w_j. \quad (5.42)$$

For example $\Psi_{qq} = \partial_q(\rho \Psi_\rho) = \rho \Psi_\rho + \rho^2 \Psi_{\rho\rho}$: again the first-derivative term is necessary. Writing $d_q = \nabla_\gamma q$, $d_a = \nabla_\gamma a$, $Q_q = \nabla_\gamma^2 q$, $Q_a = \nabla_\gamma^2 a$, the same Hessian is

$$H_{\xi\gamma} = B^\top(\rho p)d_q^\top - B^\top w d_a^\top, \quad (5.43)$$

$$H_{\gamma\gamma} = \Psi_{qq}d_q d_q^\top + \Psi_{qa}(d_q d_a^\top + d_a d_q^\top) + \Psi_{aa}d_a d_a^\top + \Psi_q Q_q + \Psi_a Q_a. \quad (5.44)$$

The inputs d_q, d_a, Q_q, Q_a are exactly (5.23). Evaluating $\log(1 + e^t)$ by a stable log-sum-exp routine and computing both logistic tails avoids unnecessary overflow and loss of $1 - p$ near one; this is an evaluation change, not a likelihood change.

5.9 Gamma recurrences near the Poisson limit

For integer $y \geq 0$, iterating $\Gamma(\rho + 1) = \rho \Gamma(\rho)$ gives

$$\log \Gamma(\rho + y) - \log \Gamma(\rho) = \sum_{k=0}^{y-1} \log(\rho + k).$$

Differentiate this finite sum:

$$\psi(\rho + y) - \psi(\rho) = \sum_{k=0}^{y-1} \frac{1}{\rho + k}, \quad (5.45)$$

$$\psi_1(\rho) - \psi_1(\rho + y) = \sum_{k=0}^{y-1} \frac{1}{(\rho + k)^2}. \quad (5.46)$$

Empty sums are zero. Put $L = \sum_j \log(1 + e^{t_j})$. Substitution into (5.42) gives

$$\Psi_q = \rho L - \sum_j \sum_{k=0}^{y_j-1} \frac{\rho}{\rho + k}, \quad (5.47)$$

$$\Psi_{qq} = \rho L + \sum_j \sum_{k=0}^{y_j-1} \left\{ -\frac{\rho}{\rho + k} + \frac{\rho^2}{(\rho + k)^2} \right\} \quad (5.48)$$

$$= \rho L - \sum_j \sum_{k=0}^{y_j-1} \frac{\rho k}{(\rho + k)^2}. \quad (5.49)$$

The last expression avoids subtracting nearly equal special functions and avoids cancellation inside each bracket. It does not promise accurate dispersion fitting at arbitrarily small α . The implementation uses these finite recurrences for ordinary integer counts up to 4096 and general special-function formulas otherwise. The cutoff is a computational choice, not a theorem.

5.10 Joint dispersion blocks for the aggregated model

The same log coordinates make nuisance derivatives explicit. Let $\delta = \log \alpha$ for one group, while keeping all regression coefficients free. From (5.23),

$$q_\delta = -1, \quad a_\delta = -1, \quad q_{\delta\delta} = a_{\delta\delta} = 0, \quad \partial_\delta d_q = \partial_\delta d_a = 0.$$

At fixed ξ, γ , differentiating $\Psi(q, a, \eta)$ gives

$$\Psi_\delta = -\Psi_q - \Psi_a, \quad U_\delta = \Psi_q + \Psi_a, \quad (5.50)$$

$$H_{\delta\delta} = \Psi_{qq} + 2\Psi_{qa} + \Psi_{aa}, \quad (5.51)$$

$$H_{\xi\delta} = -B^\top(\rho p) + B^\top w, \quad (5.52)$$

$$H_{\gamma\delta} = -d_q(\Psi_{qq} + \Psi_{qa}) - d_a(\Psi_{qa} + \Psi_{aa}). \quad (5.53)$$

These blocks, together with (5.44), form the joint observed information. They are not present in an inference calculation that holds fitted dispersion fixed. For one shared δ , add its contributions over groups; for separate group dispersions, place each block in its own nuisance column. Unlike the correctly specified independent-cell NB case, regression and shape are not generally orthogonal in expected information for this coefficient-dependent effective shape.

5.11 Poisson limit and what information is lost

At fixed v_i, u_j , $\alpha \downarrow 0$ implies $\rho = s_1 A \rightarrow \infty$ and $p_j = u_j/(u_j + A) \rightarrow 0$, with $\rho p_j \rightarrow s_1 u_j = m_j$. Then

$$w_j = (\rho + y_j)p_j(1 - p_j) \rightarrow m_j.$$

The distribution tends to $\text{Poisson}(m_j)$ for each group-voxel total. The limit does not recover the study-allocation likelihood (3.12). In particular, global covariate information can still differ from that of full Poisson CBMR.

5.12 What unrestricted SV aggregation would require

The independent-cell and clustered-NB SV formulas do not require a common spatial map. If one nevertheless wishes to moment-match SV totals over a fixed study set, use the more general quantities

$$m_j = \sum_i \mu_{ij}, \quad v_j = \sum_i \alpha_i \mu_{ij}^2, \quad \rho_j = m_j^2/v_j, \quad \kappa_j = \log m_j.$$

Here v_j denotes the *extra variance*, not the GC study factor v_i used above. Assume fixed $\alpha_i > 0$. Now ρ_j is generally voxel-dependent and both m_j and ρ_j depend on the spatial coefficients. With $x_{ij} = \nabla_\theta \eta_{ij}$, compute

$$\nabla m_j = \sum_i \mu_{ij} x_{ij}, \quad \nabla^2 m_j = \sum_i \mu_{ij} x_{ij} x_{ij}^\top, \quad (5.54)$$

$$\nabla v_j = 2 \sum_i \alpha_i \mu_{ij}^2 x_{ij}, \quad \nabla^2 v_j = 4 \sum_i \alpha_i \mu_{ij}^2 x_{ij} x_{ij}^\top. \quad (5.55)$$

For $h = m_j$ or v_j , define $L_h = \nabla h/h$ and $Q_h = \nabla^2 h/h - (\nabla h)(\nabla h)^\top/h^2$. Then

$$\nabla \kappa_j = L_m, \quad \nabla^2 \kappa_j = Q_m, \quad (5.56)$$

$$\nabla \rho_j = \rho_j(2L_m - L_v), \quad (5.57)$$

$$\nabla^2 \rho_j = \rho_j\{(2L_m - L_v)(2L_m - L_v)^\top + 2Q_m - Q_v\}. \quad (5.58)$$

Let $F(\kappa, \rho; y)$ be the negative of the independent NB scalar log-likelihood, now with *aggregate* mean $m = e^\kappa$ and shape ρ . Its partials follow from Sections 4 and 2:

$$F_\kappa = (\rho + y)m/(\rho + m) - y, \quad (5.59)$$

$$F_{\kappa\kappa} = (\rho + y)\rho m/(\rho + m)^2, \quad (5.60)$$

$$F_{\kappa\rho} = m(m - y)/(\rho + m)^2, \quad (5.61)$$

$$F_\rho = -\psi(y + \rho) + \psi(\rho) - \log \rho - 1 + \log(\rho + m) + (\rho + y)/(\rho + m), \quad (5.62)$$

$$F_{\rho\rho} = -\psi_1(y + \rho) + \psi_1(\rho) - 1/\rho + 1/(\rho + m) + (m - y)/(\rho + m)^2. \quad (5.63)$$

The complete information contribution for voxel j is

$$H_j = F_{\kappa\kappa} \nabla \kappa_j \nabla \kappa_j^\top + F_{\kappa\rho} (\nabla \kappa_j \nabla \rho_j^\top + \nabla \rho_j \nabla \kappa_j^\top) \quad (5.64)$$

$$+ F_{\rho\rho} \nabla \rho_j \nabla \rho_j^\top + F_\kappa \nabla^2 \kappa_j + F_\rho \nabla^2 \rho_j. \quad (5.65)$$

Summing over totals gives the exact Hessian of this *new working aggregated likelihood*; the moment matching itself is still approximate. This derivation does not claim that the current API implements this generalization. In particular, simply applying the grouped fixed- ρ , A spatial weight to an SV design would omit the dependencies explicitly displayed in (5.65).

6 Clustered NB: gamma integration and diagonal-minus-rank-one curvature

6.1 Integrating the shared study effect

For each independent study let $\Lambda_i \sim \text{Gamma}(\nu, \text{rate } \nu)$, so $E \Lambda_i = 1$ and $\text{Var } \Lambda_i = 1/\nu = \alpha$. Conditional on $\Lambda_i = \lambda$, cells are independent Poisson: $Y_{ij} | \lambda \sim \text{Poisson}(\lambda \mu_{ij})$. Let $n_i = \sum_j y_{ij}$ and $t_i = \sum_j \mu_{ij}$. The conditional mass and gamma density multiply to

$$p(y_i | \lambda) f(\lambda) = \left\{ \prod_j \frac{\mu_{ij}^{y_{ij}}}{y_{ij}!} \right\} e^{-\lambda t_i} \lambda^{n_i} \frac{\nu^\nu}{\Gamma(\nu)} \lambda^{\nu-1} e^{-\nu \lambda} \quad (6.1)$$

$$= \frac{\nu^\nu}{\Gamma(\nu)} \left\{ \prod_j \frac{\mu_{ij}^{y_{ij}}}{y_{ij}!} \right\} \lambda^{n_i + \nu - 1} e^{-(t_i + \nu)\lambda}. \quad (6.2)$$

Substitute $s = (t_i + \nu)\lambda$ in the integral over $\lambda > 0$:

$$\int_0^\infty \lambda^{n_i + \nu - 1} e^{-(t_i + \nu)\lambda} d\lambda = \frac{\Gamma(n_i + \nu)}{(t_i + \nu)^{n_i + \nu}}.$$

Hence the exact marginal log-likelihood is

$$\ell_i = \log \Gamma(n_i + \nu) - \log \Gamma(\nu) + \nu \log \nu - (n_i + \nu) \log(t_i + \nu) + \sum_j y_{ij} \log \mu_{ij} - \sum_j \log \Gamma(y_{ij} + 1). \quad (6.3)$$

This integrates one *shared* multiplier; multiplying independent cell NB masses would describe a different model.

6.2 Covariance and total/allocation factorization

The laws of total variance and covariance give

$$\text{Var}(Y_{ij}) = \text{E}(\Lambda_i \mu_{ij}) + \text{Var}(\Lambda_i \mu_{ij}) = \mu_{ij} + \alpha \mu_{ij}^2, \quad (6.4)$$

$$\text{Cov}(Y_{ij}, Y_{ik}) = \alpha \mu_{ij} \mu_{ik} \quad (j \neq k), \quad (6.5)$$

$$\text{Cov}(Y_i) = \text{diag}(\mu_i) + \alpha \mu_i \mu_i^\top. \quad (6.6)$$

The total has NB mean t_i and shape ν , while $Y_i \mid n_i$ is multinomial with probabilities μ_{ij}/t_i . Indeed, multiplying the NB total mass by the multinomial allocation mass cancels $n_i!$ and $t_i^{n_i}$ and reproduces the integrated mass above. For GC $\mu_{ij} = u_{gj}v_i$, these probabilities are $u_{gj}/\sum_k u_{gk}$, so global moderators affect the total but not the conditional spatial allocation. For SV that allocation can depend on the study covariates.

6.3 First and second derivatives in the full log-mean vector

At fixed ν , $\partial t_i / \partial \eta_{ij} = \mu_{ij}$. Differentiate (6.3):

$$s_{ij} = \frac{\partial \ell_i}{\partial \eta_{ij}} = y_{ij} - \frac{n_i + \nu}{t_i + \nu} \mu_{ij}. \quad (6.7)$$

Write $c_i = (n_i + \nu)/(t_i + \nu)$. Since $\partial_{\eta_{ik}} c_i = -(n_i + \nu) \mu_{ik} / (t_i + \nu)^2$,

$$\frac{\partial s_{ij}}{\partial \eta_{ik}} = -c_i \mu_{ij} \mathbf{1}\{j = k\} + \frac{n_i + \nu}{(t_i + \nu)^2} \mu_{ij} \mu_{ik}. \quad (6.8)$$

Negating produces

$$W_i^{\text{obs}} = c_i \left\{ \text{diag}(\mu_i) - \frac{\mu_i \mu_i^\top}{t_i + \nu} \right\}. \quad (6.9)$$

Because $\text{E } n_i = t_i$, $\text{E } c_i = 1$, and

$$W_i^{\text{F}} = \text{diag}(\mu_i) - \frac{\mu_i \mu_i^\top}{t_i + \nu}. \quad (6.10)$$

The subtraction is a negative rank-one correction to a diagonal matrix. It must not be replaced by independent NB diagonal weights.

6.4 An independent check using the variance of the study score

For this subsection put $D_i = \text{diag}(\mu_i)$ and $\Sigma_i = D_i + \mu_i \mu_i^\top / \nu$. Since $n_i - t_i = \mathbf{1}^\top (Y_i - \mu_i)$, rewrite (6.7) as

$$s_i = Y_i - \mu_i - \frac{\mu_i (n_i - t_i)}{t_i + \nu} \quad (6.11)$$

$$= A_i (Y_i - \mu_i), \quad A_i = I_N - \frac{\mu_i \mathbf{1}^\top}{t_i + \nu}. \quad (6.12)$$

Using $\mathbf{1}^\top D_i = \mu_i^\top$ and $\mathbf{1}^\top \mu_i = t_i$, multiply out:

$$A_i \Sigma_i = D_i + \frac{\mu_i \mu_i^\top}{\nu} - \frac{\mu_i \mu_i^\top}{t_i + \nu} - \frac{t_i \mu_i \mu_i^\top}{\nu(t_i + \nu)} \quad (6.13)$$

$$= D_i + \left\{ \frac{1}{\nu} - \frac{1}{t_i + \nu} - \frac{t_i}{\nu(t_i + \nu)} \right\} \mu_i \mu_i^\top = D_i. \quad (6.14)$$

Therefore

$$\text{Var}(s_i) = A_i \Sigma_i A_i^\top = D_i A_i^\top = D_i - \frac{\mu_i \mu_i^\top}{t_i + \nu} = W_i^{\text{F}}.$$

After projection, $\text{Var}(X_i^\top s_i) = X_i^\top W_i^{\text{F}} X_i$, as required for the correctly specified study likelihood. Ignoring the off-diagonal entries of Σ_i would not reproduce this calculation.

6.5 Why subtracting that rank-one term preserves positivity

For any vector $a \in \mathbb{R}^N$ let $S = \sum_j \mu_{ij} a_j^2$ and $b = \sum_j \mu_{ij} a_j$. Then

$$(t_i + \nu) a^\top W_i^F a = (t_i + \nu) S - b^2 \quad (6.15)$$

$$= \nu S + \left(\sum_j \mu_{ij} \right) \left(\sum_j \mu_{ij} a_j^2 \right) - \left(\sum_j \mu_{ij} a_j \right)^2 \quad (6.16)$$

$$= \nu \sum_j \mu_{ij} a_j^2 + \sum_{j < k} \mu_{ij} \mu_{ik} (a_j - a_k)^2. \quad (6.17)$$

The last identity pairs the j, k and k, j terms in the double sums. For positive ν, μ_{ij} it is strictly positive for nonzero a . Also $c_i > 0$, so the observed matrix is positive definite. After projection the regression information is positive definite when the stacked affine design has full column rank. Joint dispersion convexity is not implied.

6.6 Projection for unrestricted SV and hybrid designs

Define the projected mean vector $b_i = X_i^\top \mu_i$ (not a spatial coefficient map). By (2.14),

$$U_i = X_i^\top y_i - c_i b_i, \quad (6.18)$$

$$H_i = c_i \left\{ X_i^\top \text{diag}(\mu_i) X_i - \frac{b_i b_i^\top}{t_i + \nu} \right\}. \quad (6.19)$$

For pure SV let $a_i = B^\top \mu_i$. The (r, r') block is

$$H_{\beta_r, \beta_{r'}} = \sum_i c_i Z_{ir} Z_{ir'}^\top \left\{ B^\top \text{diag}(\mu_i) B - \frac{a_i a_i^\top}{t_i + \nu} \right\}. \quad (6.20)$$

The global and mixed blocks of a hybrid are obtained with its corresponding X_i in (6.19). These formulas do not assume separable SV means and do not allocate an $N \times N$ weight matrix.

6.7 GC reduction: differentiate the scalar spatial total twice

Within group g , use u_j, v_i as before and define

$$A^{(u)} = \sum_j u_j, \quad a = B^\top u, \quad K = B^\top \text{diag}(u) B, \quad d_i = A^{(u)} v_i + \nu.$$

At fixed global coefficients the nonlinear part of $-\ell_g$ is $f(A^{(u)}) = \sum_i (n_i + \nu) \log(A^{(u)} v_i + \nu)$. Thus

$$f' = \sum_i \frac{(n_i + \nu) v_i}{d_i}, \quad f'' = - \sum_i \frac{(n_i + \nu) v_i^2}{d_i^2}.$$

Since $\nabla_\xi A^{(u)} = a$ and $\nabla_\xi^2 A^{(u)} = K$,

$$\nabla_\xi(-\ell_g) = f' a - B^\top Y_g, \quad (6.21)$$

$$H_{\xi\xi} = f' K + f'' a a^\top. \quad (6.22)$$

For a global coordinate, differentiating each term in f' gives

$$\partial_{\gamma_q} \frac{(n_i + \nu) v_i}{A^{(u)} v_i + \nu} = (n_i + \nu) \frac{v_i z_{iq} d_i - A^{(u)} v_i^2 z_{iq}}{d_i^2} = \frac{(n_i + \nu) \nu v_i}{d_i^2} z_{iq}.$$

Consequently

$$H_{\xi\gamma} = a \left\{ \sum_i \frac{(n_i + \nu)\nu v_i}{d_i^2} z_i^\top \right\}. \quad (6.23)$$

The global negative-log-likelihood score is $\sum_i \{(n_i + \nu)A^{(u)}v_i/d_i - n_i\}z_i$. One more derivative yields

$$H_{\gamma\gamma}^{(g)} = \sum_i \frac{(n_i + \nu)\nu A^{(u)}v_i}{d_i^2} z_i z_i^\top. \quad (6.24)$$

Sum this last block over groups, while keeping each group spatial block and border. Expected information follows by replacing n_i by t_i , since each weight is affine in n_i .

6.8 Nuisance derivatives and limiting cases

The ν -dependent terms of (6.3) are exactly those of an NB total with mean t_i . Hence, at fixed μ_i ,

$$\ell_{i,\nu} = \psi(n_i + \nu) - \psi(\nu) + \log \nu + 1 - \log(t_i + \nu) - \frac{n_i + \nu}{t_i + \nu}, \quad (6.25)$$

$$\ell_{i,\nu\nu} = \psi_1(n_i + \nu) - \psi_1(\nu) + 1/\nu - 1/(t_i + \nu) - \frac{t_i - n_i}{(t_i + \nu)^2}, \quad (6.26)$$

$$\ell_{i,\eta_j\nu} = \frac{\mu_{ij}(n_i - t_i)}{(t_i + \nu)^2}. \quad (6.27)$$

Projection supplies the regression–shape cross block, and $\nu = e^{-\delta}$ gives the same log-dispersion chain rule as before. As $\alpha \downarrow 0$, $\nu \rightarrow \infty$, $c_i \rightarrow 1$, and the rank-one correction tends to zero, recovering full independent Poisson information. With one voxel, the clustered model reduces to the independent NB scalar model: (6.9) becomes $(n_i + \nu)\nu\mu_i/(\mu_i + \nu)^2$.

7 Small calculations for checking the Hessian by hand

7.1 The same covariate under GC and SV

Take one group with two studies, moderator values $a_1 = 0, a_2 = 1$, two voxels, and the saturated basis $B = I_2$. Set all coefficients and offsets to zero so every $\mu_{ij} = 1$. For GC write the single group’s basis coefficients as $\xi = (\xi_{(1)}, \xi_{(2)})^\top$ and order parameters as $(\xi_{(1)}, \xi_{(2)}, \gamma)$. The four cell design rows (study slowest) are

$$X_{\text{GC}} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{pmatrix}.$$

Poisson weights are all one, giving

$$H_{\text{GC}} = X_{\text{GC}}^\top X_{\text{GC}} = \begin{pmatrix} 2 & 0 & 1 \\ 0 & 2 & 1 \\ 1 & 1 & 2 \end{pmatrix}.$$

For SV use spatial intercept and moderator maps, ordered $(\beta_{0,1}, \beta_{0,2}, \beta_{a,1}, \beta_{a,2})$. With $Z = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}$,

$$X_{\text{SV}} = Z \otimes I_2 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{pmatrix}, \quad H_{\text{SV}} = \begin{pmatrix} 2 & 0 & 1 & 0 \\ 0 & 2 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{pmatrix}.$$

The matrices differ because there is one global moderator coefficient in GC and two voxel-specific moderator coefficients in SV. Restricting $\beta_{a,1} = \beta_{a,2} = \gamma$ gives a linear map $\beta = J\theta$; direct multiplication verifies $H_{GC} = J^\top H_{SV} J$.

7.2 Observed NB information is not Fisher information

Use the GC design above with count vector $(0, 1, 2, 3)^\top$ and $\alpha = 1$, still evaluated at $\mu = 1$. The independent NB weights are $w^{\text{obs}} = (1, 2, 3, 4)^\top/4$, whereas $w^F = \mathbf{1}/2$. Weighted cross-products give

$$H_{\text{NB}} = \begin{pmatrix} 1 & 0 & 3/4 \\ 0 & 3/2 & 1 \\ 3/4 & 1 & 7/4 \end{pmatrix}, \quad I_{\text{NB}} = \begin{pmatrix} 1 & 0 & 1/2 \\ 0 & 1 & 1/2 \\ 1/2 & 1/2 & 1 \end{pmatrix}.$$

For instance the $(1, 3)$ observed entry is contributed only by study 2, voxel 1: its design product is one and its weight is $(1 + 2)/4 = 3/4$. Every other entry can be checked by the same four-row sum. These are evaluations at a specified parameter, not claims about a fitted MLE.

7.3 Clustered curvature for a two-voxel study

Let $\mu_i = (1, 2)^\top$, $\nu = 2$, and $n_i = 4$. Then $t_i = 3$, $c_i = 6/5$, and

$$W_i^F = \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix} - \frac{1}{5} \begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix} = \begin{pmatrix} 4/5 & -2/5 \\ -2/5 & 6/5 \end{pmatrix}, \quad W_i^{\text{obs}} = \frac{6}{5} W_i^F.$$

The off-diagonal entries are nonzero even though cells were conditionally independent before integrating the shared gamma effect. The Fisher determinant is $20/25 > 0$.

7.4 Unequal study factors in moment matching

Take one group, two studies with $v = (1, 2)$, $\alpha = 1/2$, and a voxel with $u = 3$. Then $s_1 = 3$, $s_2 = 5$, $s_3 = 9$, $A = 6/5$, $\rho = 18/5$, $m = 9$, and

$$\text{Var}(Y_{\cdot j}) = 9 + \frac{1}{2} 9 \cdot 5 = \frac{63}{2}, \quad m + m^2/\rho = 9 + 81/(18/5) = \frac{63}{2}.$$

The matched probability is $p = 3/(3 + 6/5) = 5/7$. For $y = 4$ the spatial log-mean observed weight is $(\rho + y)p(1 - p) = (38/5)(5/7)(2/7) = 76/49$. The third-cumulant discrepancy is $2(1/2)^2 3^3(9 - 25/3) = 9$, demonstrating that matching the first two moments did not match the distribution.

8 From group formulas to shared spatial patterns

8.1 Patterns are fixed by the design, not the fitted intensities

The implementation can share calculations between studies with identical spatial-design rows even when those rows are not disjoint group indicators. Let $h = 1, \dots, K_{\text{pat}}$ index distinct rows, and let $L_h \in \mathbb{R}^R$ be the spatial loading column for pattern h . If study i has that pattern, its spatial part is

$$B_j^\top c_h, \quad c_h = \sum_{r=1}^R L_{hr} \beta_r = J_h \beta, \quad J_h = L_h^\top \otimes I_P.$$

Here J_h has dimension $P \times RP$. Two studies share a pattern because their spatial loadings agree for every parameter value, not merely because two fitted maps happen to agree at the current iterate. With a study exposure e_i , the mean is $\mu_{ij} = \exp(B_j^\top c_h) e_i \exp(z_i^\top \gamma)$ within that pattern.

For Poisson, (3.6) can be applied to each pattern, retaining pattern-voxel counts and study totals. For clustered NB, the same grouping of its exact study likelihood is valid. For aggregated

NB, moment matching is carried out within each pattern; this remains a working likelihood, subject to the qualifications above. The group formulas are the special case $L_h = e_h$.

8.2 The scattering rule is another affine chain rule

Let a pattern's negative log-likelihood be $F_h(c_h, \gamma)$. Suppose the formulas already derived give

$$K_h = \nabla_{c_h}^2 F_h, \quad G_h = \nabla_{c_h} \nabla_\gamma^\top F_h, \quad E_h = \nabla_\gamma^2 F_h.$$

Since $c_h = J_h \beta$ with fixed J_h , there is no second derivative of the inner map. Therefore

$$\nabla_\beta F_h = J_h^\top \nabla_{c_h} F_h, \quad (8.1)$$

$$H_{\beta\beta} = \sum_h J_h^\top K_h J_h = \sum_h (L_h L_h^\top) \otimes K_h, \quad (8.2)$$

$$H_{\beta\gamma} = \sum_h J_h^\top G_h, \quad H_{\gamma\gamma} = \sum_h E_h. \quad (8.3)$$

In individual coefficient blocks, $H_{\beta_r \beta_{r'}} = \sum_h L_{hr} L_{hr'} K_h$ and $H_{\beta_r \gamma} = \sum_h L_{hr} G_h$. This explains why one weighted basis product per pattern can feed several blocks without differentiating the large prediction array. If a pattern loads two spatial terms, it contributes to their cross-information: it must not be treated as an independently fitted group simply because its likelihood was evaluated in a separate loop.

8.3 Computational improvement and its exact boundary

For independent-cell likelihoods, a weighted basis product is

$$K = B^\top \text{diag}(w) B = B^\top \tilde{B}, \quad \tilde{B}_{jp} = w_j B_{jp}.$$

The right-hand construction multiplies every row of B by its weight; it does not construct the diagonal matrix. For a dense basis its cost is $O(NP^2)$ and its weighted-basis workspace is $O(NP)$, in addition to storing the $P \times P$ result. The clustered formula adds a P -vector and its outer product:

$$B^\top \left\{ c \text{diag}(\mu) - a \mu \mu^\top \right\} B = c B^\top \text{diag}(\mu) B - a (B^\top \mu) (B^\top \mu)^\top.$$

There is no need to form an $N \times N$ matrix.

The full-cell SV formula (2.13) similarly costs $O(MNP^2 + MR^2P^2)$ for straightforward dense studywise assembly, with $O((RP)^2)$ storage if a dense information matrix is requested. Identical design rows permit summing the relevant weights before forming each basis product. Sparse bases and sparse loading supports can reduce this cost further, but the exact reduction depends on their actual sparsity, not merely their nominal dimensions.

These formulas avoid the parameter-by-pattern-by-voxel derivative intermediate used by the source automatic-differentiation strategy. They do not make an arbitrary $n_\theta \times n_\theta$ covariance free to store: materializing a dense covariance still needs $O(n_\theta^2)$ entries. The following covariance chapters derive how to avoid that allocation when only variances or a small number of contrasts are needed.

9 Design zeros, information blocks, and what an inverse means

9.1 Dimensions and the statistical object being inverted

Throughout this fragment, there are M studies, N voxels, P spatial basis coefficients per map, G groups, R spatially varying (SV) covariates, and Q global covariates. The basis matrix is

$B \in \mathbb{R}^{N \times P}$, with row j written $B_j^\top$: $B_j \in \mathbb{R}^P$ is a column vector. Likewise $Z_i \in \mathbb{R}^R$ and $z_i \in \mathbb{R}^Q$ are column vectors of study covariates. The coefficient vector θ contains regression coefficients only unless dispersion is explicitly added in Section 14.

For SV maps, write $\beta = (\beta_1^\top, \dots, \beta_R^\top)^\top \in \mathbb{R}^{RP}$. For group maps, use $\beta = (\xi_1^\top, \dots, \xi_G^\top)^\top \in \mathbb{R}^{GP}$. Thus the symbol β always denotes stacked spatial coefficients, but its dimension depends on the specification. With global coefficients, $\theta = (\beta^\top, \gamma^\top)^\top$ and $\gamma \in \mathbb{R}^Q$. Let $d = \dim(\beta)$ and $n_\theta = d + Q$; set $Q = 0$ when no global coefficients are fitted. Typical affine predictors, with a known offset o_{ij} , are

$$\eta_{ij} = o_{ij} + (Z_i^\top \otimes B_j^\top)\beta + z_i^\top \gamma \quad (\text{SV specification}), \quad (9.1)$$

$$\eta_{ij} = o_{ij} + B_j^\top \xi_{g(i)} + z_i^\top \gamma \quad (\text{group specification}). \quad (9.2)$$

An offset is not an additional estimated coefficient.

Our convention is

$$H(\theta) = -\nabla_\theta^2 \ell(\theta), \quad I(\theta) = \mathbb{E}_\theta\{H(\theta)\}. \quad (9.3)$$

Here H is *observed* information. Under a correctly specified regular likelihood, interchange of differentiation and integration also gives $I = \text{Var}_\theta\{\nabla_\theta \ell\}$. That information equality is a statistical assertion with hypotheses, not a consequence of matrix symmetry. Dispersion α is fixed for now.

Identities below about H^{-1} are exact finite-dimensional algebra when the stated inverses exist. Interpreting $H(\hat{\theta})^{-1}$ or $I(\hat{\theta})^{-1}$ as a sampling covariance is generally a first-order likelihood approximation, requiring identification, a suitable independent-unit asymptotic regime, and an interior regular fit. An indefinite observed information matrix may be invertible without its inverse being a valid covariance. These two questions must remain separate.

9.2 From coefficient support to a graph of SV blocks

For an independent-cell log likelihood, put $w_{ij} = -\partial^2 \ell_{ij} / \partial \eta_{ij}^2$. The affine-design chain rule, rather than any particular count family, gives the $P \times P$ spatial block

$$H_{rr'} = \sum_{i=1}^M \sum_{j=1}^N w_{ij} Z_{ir} Z_{ir'} B_j B_j^\top. \quad (9.4)$$

If $Z_{ir} Z_{ir'} = 0$ for every study i , every summand vanishes, irrespective of the numerical weights. Consequently $H_{rr'} = 0$. Conversely, a zero block at one parameter value might arise from cancellation of nonzero terms. Such an accidental zero is not a design-guaranteed decomposition that can safely be reused at other fits.

Construct a graph with one vertex for each SV covariate column. Connect r and r' whenever at least one study has both $Z_{ir} \neq 0$ and $Z_{ir'} \neq 0$. If two vertices lie in different connected components, there is no such study, and their entire information cross block vanishes. If component a has r_a columns, it contributes $d_a = r_a P$ coefficients. After permuting coefficients by components,

$$D = \text{blkdiag}(D_1, \dots, D_{K_{\text{comp}}}), \quad \sum_{a=1}^{K_{\text{comp}}} d_a = d. \quad (9.5)$$

For a group-only spatial design, these are precisely the G group blocks, so we write $D = \text{blkdiag}(D_g : g = 1, \dots, G)$. For a general SV design, component labels a replace group labels g ; a component need not be a scientific group.

A path is enough to prevent independent inversion. For example, suppose column 1 co-occurs with column 2, and column 2 with column 3, while columns 1 and 3 never co-occur. The

information can have the form

$$\begin{pmatrix} a & b & 0 \\ b & c & e \\ 0 & e & f \end{pmatrix}.$$

When it is nonsingular, its $(1,3)$ inverse entry is $be/\det(H)$, which is generally nonzero. A missing direct edge does not imply a zero inverse entry. Disjoint cell-means indicators separate; a common SV intercept or another overlapping column can connect them. An all-zero design column instead produces an unidentified direction.

For a likelihood factorized by study but not by voxel, a study contribution may couple its voxels. Nevertheless, it cannot differentiate with respect to two spatial columns that are not both present in that study, so the same support argument applies when that factorization and predictor support hold. If an aggregate likelihood couples studies or groups, its actual factor supports must be inspected instead. Similarly, a penalty that couples components adds edges; design zeros for the likelihood alone are not then zeros for the penalized objective.

9.3 Verification of independent block inversion

First suppose there is no global border. Let Π be the permutation matrix putting the components in order, and suppose each D_a is nonsingular. Define

$$W = \Pi^\top \text{blkdiag}(D_1^{-1}, \dots, D_{K_{\text{comp}}}^{-1})\Pi, \quad H = \Pi^\top D\Pi.$$

Since $\Pi\Pi^\top = I_d$, multiplication gives

$$HW = \Pi^\top \text{blkdiag}(D_1 D_1^{-1}, \dots, D_{K_{\text{comp}}} D_{K_{\text{comp}}}^{-1})\Pi = \Pi^\top I_d \Pi = I_d.$$

Multiplying in the reverse order gives $WH = I_d$ as well. Thus $W = H^{-1}$. This is an exact identity, not an approximation obtained by ignoring small cross blocks. It also shows why parameter ordering must accompany a collection of blocks: block coordinates and original coefficient coordinates need not coincide.

With shared global coefficients, the full information instead has the form

$$H = \begin{pmatrix} D & F \\ F^\top & \mathcal{E} \end{pmatrix}, \quad F \in \mathbb{R}^{d \times Q}, \quad \mathcal{E} \in \mathbb{R}^{Q \times Q}. \quad (9.6)$$

Here $\mathcal{E}$ denotes a matrix, not an expectation or an intensity total. Although D has disconnected spatial blocks, paths through the global coefficients connect the full system.

10 Schur elimination and the covariance induced by shared coefficients

10.1 Deriving a solve one equation at a time

Assume D is nonsingular. For a right-hand side $b = (b_D^\top, b_\gamma^\top)^\top$, write the desired solution as $x = (x_D^\top, x_\gamma^\top)^\top$. The two block equations in $Hx = b$ are

$$Dx_D + Fx_\gamma = b_D, \quad (10.1)$$

$$F^\top x_D + \mathcal{E}x_\gamma = b_\gamma. \quad (10.2)$$

Solving the first equation for the spatial part gives $x_D = D^{-1}b_D - D^{-1}Fx_\gamma$. Substitute this expression into the second equation:

$$F^\top D^{-1}b_D - F^\top D^{-1}Fx_\gamma + \mathcal{E}x_\gamma = b_\gamma, \quad (10.3)$$

$$(\mathcal{E} - F^\top D^{-1}F)x_\gamma = b_\gamma - F^\top D^{-1}b_D. \quad (10.4)$$

Define, once for all right-hand sides,

$$T = D^{-1}F, \quad S = \mathcal{E} - F^\top T. \quad (10.5)$$

If S is nonsingular, the complete solve is therefore

$$y = D^{-1}b_D, \quad (10.6)$$

$$x_\gamma = S^{-1}(b_\gamma - F^\top y), \quad (10.7)$$

$$x_D = y - Tx_\gamma. \quad (10.8)$$

The notation $D^{-1}b_D$ means a solve with the factorized blocks of D ; it is not an instruction to construct a dense inverse. These equations also hold for several right-hand sides stored as columns.

10.2 Full inverse and explicit multiplication check

Expanding the solution in its two right-hand-side components gives

$$x_D = (D^{-1} + TS^{-1}T^\top)b_D - TS^{-1}b_\gamma, \quad x_\gamma = -S^{-1}T^\top b_D + S^{-1}b_\gamma.$$

We used symmetry of D to obtain $F^\top D^{-1} = T^\top$. Hence

$$H^{-1} = \begin{pmatrix} D^{-1} + TS^{-1}T^\top & -TS^{-1} \\ -S^{-1}T^\top & S^{-1} \end{pmatrix}. \quad (10.9)$$

For a direct audit, multiply H by the proposed inverse. Its four blocks simplify separately, using $DT = F$:

$$(HH^{-1})_{DD} = I_d + DTS^{-1}T^\top - FS^{-1}T^\top = I_d, \quad (10.10)$$

$$(HH^{-1})_{D\gamma} = -DTS^{-1} + FS^{-1} = 0, \quad (10.11)$$

$$(HH^{-1})_{\gamma D} = T^\top + (F^\top T - \mathcal{E})S^{-1}T^\top = T^\top - SS^{-1}T^\top = 0, \quad (10.12)$$

$$(HH^{-1})_{\gamma\gamma} = (\mathcal{E} - F^\top T)S^{-1} = I_Q. \quad (10.13)$$

The transpose verifies the reverse product because both matrices are symmetric. When $F = 0$, the formula reduces to separate inverses of D and $\mathcal{E}$. When $Q = 0$, there is no Schur calculation.

Invertibility of H alone does not guarantee invertibility of D . For example, $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ is nonsingular but its leading scalar block is zero. The particular elimination above is unavailable in that case; a different pivoting scheme or full symmetric indefinite factorization is needed.

10.3 Positive definiteness by completing the square

For arbitrary $u \in \mathbb{R}^d$ and $v \in \mathbb{R}^Q$, expand

$$\begin{pmatrix} u \\ v \end{pmatrix}^\top H \begin{pmatrix} u \\ v \end{pmatrix} = u^\top Du + 2u^\top Fv + v^\top \mathcal{E}v \quad (10.14)$$

$$= (u + Tv)^\top D(u + Tv) + v^\top Sv. \quad (10.15)$$

Indeed, the cross term is $2u^\top DTv = 2u^\top Fv$ and the remaining v term is $v^\top (T^\top DT + S)v = v^\top \mathcal{E}v$. Thus, if $D \succ 0$, then $H \succ 0$ if and only if $S \succ 0$. Necessity follows by choosing $u = -Tv$; sufficiency follows because the sum is positive for any nonzero (u, v) . With $D \succ 0$, the analogous statement $H \succeq 0$ if and only if $S \succeq 0$ holds, but a singular S prevents an ordinary inverse.

Equivalently,

$$H = \begin{pmatrix} I_d & 0 \\ T^\top & I_Q \end{pmatrix} \begin{pmatrix} D & 0 \\ 0 & S \end{pmatrix} \begin{pmatrix} I_d & T \\ 0 & I_Q \end{pmatrix}. \quad (10.16)$$

This is a congruence, not an orthogonal diagonalization. It determines inertia and gives $\det(H) = \det(D)\det(S)$, but it does not say that the eigenvalues of D and S are the eigenvalues of H .

10.4 Joint versus fixed-parameter covariance and profile curvature

Suppose $H \succ 0$ and use $V = H^{-1}$ as the local model-based covariance approximation. If γ were known rather than estimated, the spatial curvature would be D and its covariance approximation would be D^{-1} . Joint estimation instead gives

$$V_{\beta\beta} = D^{-1} + TS^{-1}T^\top, \quad V_{\beta\gamma} = -TS^{-1}, \quad V_{\gamma\gamma} = S^{-1}. \quad (10.17)$$

For any $a \in \mathbb{R}^d$, $a^\top TS^{-1}T^\top a = (T^\top a)^\top S^{-1}(T^\top a) \geq 0$. Thus the extra joint-estimation term is positive semidefinite (PSD). This statement is about inverse curvature; it is not a general ordering theorem for arbitrary robust covariance estimators.

In a Gaussian approximation with precision H , completing the square also shows that the conditional covariance of β given γ is D^{-1} . This is the precise local meaning of “conditional” here; it does not assert an exact conditional sampling law for an arbitrary finite-sample estimator. Similarly, $\mathcal{E}^{-1}$ is the local covariance for γ if β were known. Joint inference uses S^{-1} instead. Since $\mathcal{E} - S = F^\top D^{-1}F \succeq 0$, both matrices being SPD implies $S^{-1} \succeq \mathcal{E}^{-1}$.

The Schur complement also has an optimization interpretation. Let $L = -\ell$, and let $\beta_*(\gamma)$ be a differentiable local minimizer in β , satisfying $L_\beta(\beta_*(\gamma), \gamma) = 0$. Implicit differentiation gives

$$D \frac{\partial \beta_*}{\partial \gamma^\top} + F = 0, \quad \frac{\partial \beta_*}{\partial \gamma^\top} = -D^{-1}F = -T.$$

For the profile objective $L_{\text{prof}}(\gamma) = L(\beta_*(\gamma), \gamma)$, the term involving L_β vanishes:

$$\nabla_\gamma L_{\text{prof}} = L_\gamma, \quad \nabla_\gamma^2 L_{\text{prof}} = \mathcal{E} + F^\top \frac{\partial \beta_*}{\partial \gamma^\top} = \mathcal{E} - F^\top D^{-1}F = S.$$

These derivatives are exact along a smooth stationary branch. Treating this branch as a local profile minimum additionally requires appropriate curvature; global uniqueness does not follow from the calculation.

10.5 Cross-group covariance and a hand-computable example

For group blocks let $T_g = D_g^{-1}F_g$, so T stacks the T_g . Equation (10.17) gives

$$V_{\xi_g \xi_h} = \mathbf{1}\{g = h\}D_g^{-1} + T_g S^{-1}T_h^\top. \quad (10.18)$$

The entire spatial correction $TS^{-1}T^\top$ has rank at most Q . Each cross-group block has rank at most Q as well, though its entries can be positive or negative; an off-diagonal rectangular block is not itself a PSD matrix. A small number of global coefficients therefore induces low-rank, not necessarily negligible, dependence among many maps.

Take $G = 2$, $P = Q = 1$, and choose

$$D = \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix}, \quad F = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad \mathcal{E} = 2.$$

The arithmetic in the elimination is

$$T = \begin{pmatrix} 1/2 \\ 1/3 \end{pmatrix}, \quad F^\top T = \frac{1}{2} + \frac{1}{3} = \frac{5}{6}, \quad S = 2 - \frac{5}{6} = \frac{7}{6}, \quad S^{-1} = \frac{6}{7}.$$

Both D and S are positive definite, so the bordered matrix is SPD. The spatial correction is

$$TS^{-1}T^\top = \frac{6}{7} \begin{pmatrix} 1/4 & 1/6 \\ 1/6 & 1/9 \end{pmatrix} = \begin{pmatrix} 3/14 & 1/7 \\ 1/7 & 2/21 \end{pmatrix}.$$

Adding D^{-1} and the remaining blocks yields

$$H = \begin{pmatrix} 2 & 0 & 1 \\ 0 & 3 & 1 \\ 1 & 1 & 2 \end{pmatrix}, \quad H^{-1} = \frac{1}{7} \begin{pmatrix} 5 & 1 & -3 \\ 1 & 3 & -2 \\ -3 & -2 & 6 \end{pmatrix}. \quad (10.19)$$

For example, the first row of the product is $\{2(5, 1, -3) + (-3, -2, 6)\}/7 = (1, 0, 0)$; the other rows are checked in the same way. Spatial estimates have covariance $1/7$ despite a zero spatial information cross entry.

For the group difference $c^\top \theta = \xi_1 - \xi_2$,

$$c^\top H^{-1} c = \frac{5}{7} + \frac{3}{7} - 2\frac{1}{7} = \frac{6}{7}.$$

Using the correct marginal variances but discarding their covariance would give $8/7$, not $6/7$. Treating γ as known would instead give $1/2 + 1/3 = 5/6$. The joint correction for this difference is only $(1/2 - 1/3)^2(6/7) = 1/42$, and indeed $5/6 + 1/42 = 6/7$. This demonstrates both distinctions: joint versus known global effects, and joint marginals versus incorrectly independent marginals. The direction of error from dropping cross covariances need not always be the same as in this example.

11 Selected covariance calculations without a full inverse

11.1 Project by solving for the contrasts

Let $C \in \mathbb{R}^{k \times n_\theta}$ contain k fixed linear contrasts. Solving

$$HX = C^\top, \quad V_C = CX \quad (11.1)$$

gives $X = H^{-1}C^\top$ and hence $V_C = CH^{-1}C^\top$. Thus only k right-hand sides are needed, rather than n_θ columns of an inverse. Symmetry gives $V_C^\top = (CH^{-1}C^\top)^\top = V_C$ exactly; numerical symmetrization can remove roundoff asymmetry but cannot repair a wrong model or an indefinite result.

There is a useful reduced formula. Partition $C = [C_D \ C_\gamma]$ conformably with the border. Multiplication of the four inverse blocks gives

$$V_C = C_D D^{-1} C_D^\top + C_D T S^{-1} T^\top C_D^\top \quad (11.2)$$

$$- C_D T S^{-1} C_\gamma^\top - C_\gamma S^{-1} T^\top C_D^\top + C_\gamma S^{-1} C_\gamma^\top \quad (11.3)$$

$$= C_D D^{-1} C_D^\top + (C_\gamma - C_D T) S^{-1} (C_\gamma - C_D T)^\top. \quad (11.4)$$

For a spatial-only contrast, its uncertainty from the global fit vanishes exactly when $C_D T = 0$ (assuming $S \succ 0$). For a contrast involving both kinds of coefficients, cancellation instead concerns $C_\gamma - C_D T$. These statements follow from the positive definite quadratic form and are not an assumption of group independence.

Computationally, solve $DY = C_D^\top$, form $A_C = C_\gamma - C_D T$, and solve $SW = A_C^\top$. Then compute $C_D Y + A_C W$. Neither D^{-1} nor S^{-1} must be stored as a matrix. The same projection principle works with an expected information matrix or a specified penalized curvature, but that substitution changes which covariance interpretation is being used.

11.2 Diagonal entries and contrasts in batches

Let e_a denote a coordinate vector and $t_a = T^\top e_a \in \mathbb{R}^Q$. For a spatial coordinate in block g ,

$$(H^{-1})_{aa} = (D_g^{-1})_{aa} + t_a^\top S^{-1} t_a. \quad (11.5)$$

The first term means the corresponding local coordinate in D_g . To compute the second terms together, solve $SW = T^\top$; the a th correction is the inner product of row a of T with column a of W . Global diagonal entries are the diagonal of S^{-1} , obtained from solves against its coordinate vectors.

Even obtaining every diagonal entry of a dense block inverse may require many triangular solves. The benefit is that the full $d \times d$ spatial covariance is never formed. If just a few coordinates are required, solve only for their coordinate vectors. If many voxel contrasts are needed, process a manageable batch of contrast rows and retain the resulting variances or small covariance blocks. An $N \times N$ covariance of all voxels should be formed only if it is actually the requested output.

12 Independent-unit sandwich covariance and its normalization

12.1 Taylor expansion of an unnormalized score equation

Let $U_i(\theta) \in \mathbb{R}^{n_\theta}$ be the contribution of independent study i to an estimating equation, and let $U(\theta) = \sum_{i=1}^M U_i(\theta)$. At the target θ_0 assume $E\{U(\theta_0)\} = 0$; common likelihood and conditional-mean setups also have $E\{U_i(\theta_0)\} = 0$ separately. Define

$$A(\theta) = -\frac{\partial U(\theta)}{\partial \theta^\top} = -\sum_i \frac{\partial U_i(\theta)}{\partial \theta^\top}.$$

Expanding $U(\hat{\theta}) = 0$ about θ_0 gives

$$0 = U(\theta_0) - A(\theta_0)(\hat{\theta} - \theta_0) + \text{higher-order remainder}, \quad (12.1)$$

$$\hat{\theta} - \theta_0 \approx A(\theta_0)^{-1} \sum_i U_i(\theta_0). \quad (12.2)$$

This linearization requires a negligible remainder in the chosen asymptotic regime and a nonsingular limiting sensitivity. Independence between studies gives

$$\text{Var}\left\{\sum_i U_i(\theta_0)\right\} = \sum_i \text{Var}\{U_i(\theta_0)\} = J_0.$$

With appropriate laws of large numbers, a customary empirical replacement is $\hat{J} = \sum_i U_i(\hat{\theta})U_i(\hat{\theta})^\top$, leading to

$$\hat{V}_{\text{robust}} = \hat{A}^{-1} \hat{J} \hat{A}^{-\top}. \quad (12.3)$$

Here $A^{-\top} = (A^{-1})^\top$. Likelihood scores give $A = H$; general estimating equations need not give a symmetric A . Consistency of the uncentered meat requires the relevant moment conditions. In particular, systematic nonzero individual score means under a misspecified fixed design cannot simply be identified with within-study sampling variances.

There is *no additional factor* $1/M$ in Equation (12.3): its bread and meat are both unnormalized sums. To check, write $\bar{A} = \hat{A}/M$ and $\bar{J} = \hat{J}/M$. Then

$$\frac{1}{M} \bar{A}^{-1} \bar{J} \bar{A}^{-\top} = \frac{1}{M} (M \hat{A}^{-1}) (\hat{J}/M) (M \hat{A}^{-\top}) = \hat{A}^{-1} \hat{J} \hat{A}^{-\top}.$$

Mixing an averaged bread with an unscaled summed meat introduces spurious powers of M .

12.2 Why the clustering unit matters

Suppose a working likelihood is cell additive and $U_i = \sum_{j=1}^N u_{ij}$. Expanding its outer product shows exactly what study clustering retains:

$$U_i U_i^\top = \left(\sum_j u_{ij} \right) \left(\sum_l u_{il} \right)^\top \quad (12.4)$$

$$= \sum_j u_{ij} u_{ij}^\top + \sum_{j \neq l} u_{ij} u_{il}^\top. \quad (12.5)$$

A cellwise meat retains only the first sum. A study-clustered meat retains both, allowing dependence among voxels within a study. It still requires independent studies, or a justified asymptotic replacement for that assumption. If experiments share participants or another sampling unit, the independence unit may be larger than an experiment and the scores must be grouped accordingly.

More voxels within a fixed small number of studies do not automatically provide the independent replication required by cluster asymptotics. Finite-sample adjustments also do not create that replication. For example, multiplying $\hat{J}$ by $M/(M - n_\theta)$ is a particular HC1 convention, defined only for $M > n_\theta$; it is not a universal small-cluster validity theorem.

12.3 Family-specific scores cannot be interchanged

Write $x_{ij} = \nabla_\theta \eta_{ij}$ as a column vector. For the Poisson working likelihood its study score is

$$U_i^P = \sum_j x_{ij} (y_{ij} - \mu_{ij}).$$

For independent-cell NB2 with fixed dispersion α , the corresponding score is instead

$$U_i^{NB} = \sum_j x_{ij} \frac{y_{ij} - \mu_{ij}}{1 + \alpha \mu_{ij}}.$$

These formulas state the score inputs; the count-likelihood derivative derivations belong to the likelihood chapters. The NB denominators change both the estimating equation and its derivative. Combining Poisson residual scores with NB likelihood information is not the matching likelihood sandwich. Under a correctly specified conditional mean, the expected derivative of this independent-NB working score has the usual NB Fisher weights, even if the working variance is wrong; using those expected weights as bread then needs its own consistency argument. Observed NB information and expected Fisher information do not in general agree at a realized dataset.

For the SV predictor the Poisson spatial and global score pieces are explicitly

$$U_{i,\beta}^P = Z_i \otimes \left\{ \sum_j B_j (y_{ij} - \mu_{ij}) \right\}, \quad U_{i,\gamma}^P = z_i \sum_j (y_{ij} - \mu_{ij}).$$

Thus scores can be assembled by projecting residuals onto the basis, without constructing every full design row.

A study-level clustered-NB likelihood obtained by integrating a shared study effect has score $U_i = \nabla_\theta \ell_i$, including that integration-induced dependence; it is not an independent-cell NB score. An aggregate-NB likelihood for group-by-voxel totals is more problematic for study clustering: its objective need not be a sum of original-study log likelihoods at all. A matched shape parameter can depend on several studies' means simultaneously. Arbitrarily allocating an aggregate score back to studies does not supply independent score contributions, even if the allocated vectors sum to the total score. A valid sandwich must follow an actual independent-unit estimating

representation or a justified joint sampling model for the aggregates. Alternatively, a bootstrap can refit the aggregate procedure to appropriately resampled original independent units when those data and assumptions are available.

In `cbmr_improved/covariance.py`, the implemented likelihood-score sandwich explicitly requires Poisson and rejects the NB families. Its study-score path uses basis projections of counts and fitted means. The matrix machinery in `structured.py` can accept other supplied scores, but cannot certify that the caller's scores represent the fitted likelihood or the correct independence units.

12.4 Projected Gram computation and rank limits

Let $\mathbf{U} \in \mathbb{R}^{M \times n_\theta}$ have row i equal to $U_i^\top$, so $\hat{J} = \mathbf{U}^\top \mathbf{U}$. For symmetric bread H and a contrast matrix C , solve $HX = C^\top$ and put $\mathbf{V} = \mathbf{U}X$. Then

$$\mathbf{V}^\top \mathbf{V} = X^\top \mathbf{U}^\top \mathbf{U} X \quad (12.6)$$

$$= CH^{-1} \hat{J} H^{-1} C^\top = C \hat{V}_{\text{robust}} C^\top. \quad (12.7)$$

Equivalently, accumulate outer products $v_i v_i^\top$ with $v_i = CH^{-1}U_i$. For a nonsymmetric bread, the correct projected solve is $A^\top X = C^\top$, not $AX = C^\top$. This distinction disappears for the symmetric observed information.

The Gram representation proves PSD of the sandwich even if an invertible bread is indefinite. It does not make such a bread a valid likelihood covariance or prove consistency of the estimator. It also shows

$$\text{rank}(\hat{V}_{\text{robust}}) \leq \min(M, n_\theta), \quad \text{rank}(C \hat{V}_{\text{robust}} C^\top) \leq \min(M, n_\theta, k).$$

At an exactly solved unpenalized score equation, $\mathbf{U}^\top \mathbf{1}_M = \sum_i U_i = 0$, which tightens the first bound to $\min(M - 1, n_\theta)$ and similarly for the projected matrix. Explicitly centering score rows also imposes this dependence. At a penalized optimum the unpenalized score sum is generally nonzero, so the $M - 1$ argument does not automatically apply. Multiplying a Gram matrix by a positive correction factor cannot increase its rank.

13 Quadratic roughness penalties and the meaning of penalized uncertainty

13.1 Deriving the gradient and Hessian from the quadratic form

Let $J_B = J_B^\top \succeq 0$ be a fixed $P \times P$ roughness matrix and let $\lambda_g \geq 0$ be fixed smoothing parameters. For group maps use the explicit convention

$$\mathcal{P}(\theta) = \frac{1}{2} \sum_{g=1}^G \lambda_g \xi_g^\top J_B \xi_g = \frac{1}{2} \theta^\top K \theta, \quad K = \text{blkdiag}(\lambda_1 J_B, \dots, \lambda_G J_B, 0_{Q \times Q}). \quad (13.1)$$

For SV maps, replace the group blocks by $\lambda_r J_B$, $r = 1, \dots, R$. The construction of J_B depends on the specified roughness operator and basis; it is not determined by the word “smoothing” alone.

For one coefficient vector ξ , expand $\xi^\top J_B \xi = \sum_{a,b} \xi_a (J_B)_{ab} \xi_b$. Differentiation with respect to ξ_c gives

$$\frac{\partial}{\partial \xi_c} \left(\frac{\lambda}{2} \xi^\top J_B \xi \right) = \frac{\lambda}{2} \left\{ \sum_b (J_B)_{cb} \xi_b + \sum_a \xi_a (J_B)_{ac} \right\} \quad (13.2)$$

$$= \lambda (J_B \xi)_c. \quad (13.3)$$

Differentiating again gives $\partial^2 \mathcal{P} / \partial \xi_c \partial \xi_e = \lambda (J_B)_{ce}$. Equivalently, the differential is $d\mathcal{P} = (K\theta)^\top d\theta$ and its second differential is $d\theta^\top K d\theta$. A nonsymmetric supplied matrix would contribute its symmetric part; the convention here assumes symmetry.

For the minimized objective $L_{\text{pen}} = -\ell + \mathcal{P}$,

$$\nabla L_{\text{pen}} = -U + K\theta, \quad H_{\text{pen}} = H + K, \quad U(\hat{\theta}) = K\hat{\theta}. \quad (13.4)$$

Omitting the factor 1/2 in the penalty doubles both derivatives at the same numerical λ_g . Fixed separate-map penalties preserve the likelihood's spatial component zeros and global border. A between-map penalty need not preserve them.

13.2 Frequentist linearization, bias, and posterior curvature

Let θ_0 be the unpenalized population target. Expanding the penalized score equation about that target gives

$$0 \approx U(\theta_0) - H(\theta_0)(\hat{\theta} - \theta_0) - K\theta_0 - K(\hat{\theta} - \theta_0), \quad (13.5)$$

$$(H + K)(\hat{\theta} - \theta_0) \approx U(\theta_0) - K\theta_0. \quad (13.6)$$

Replacing the random sensitivity by its suitable population counterpart A_0 identifies both pieces:

$$E(\hat{\theta} - \theta_0) \approx -(A_0 + K)^{-1} K\theta_0, \quad \text{Var}(\hat{\theta}) \approx (A_0 + K)^{-1} J_0 (A_0 + K)^{-\top}.$$

The first expression describes local smoothing bias. When the penalty does not become negligible, a population penalized target can differ substantially from θ_0 , and linearization should instead be performed at that target. The displayed local bias is not a globally exact formula for a nonlinear count regression.

A plug-in penalized sandwich uses $H + K$ as bread and an appropriate estimate of the variability of the *data* score as meat. The deterministic penalty is not an independent random contribution and must not be added to every study's score. Under strong shrinkage, data-score outer products may also require attention to centering and the definition of the penalized population target; an uncentered meat is not universally consistent when individual score means are nonzero. Under a correctly specified likelihood, the local expected-curvature version is $(I + K)^{-1} I (I + K)^{-1}$.

By contrast, a Gaussian prior with precision K gives negative log-posterior curvature $H + K$. A Laplace posterior approximation uses $(H + K)^{-1}$. It is exactly the posterior covariance only for an actually quadratic posterior with appropriate support. A singular roughness precision corresponds to unpenalized directions and need not define a proper prior on its own.

The expected-curvature versions differ by the exact identity

$$(I + K)^{-1} - (I + K)^{-1} I (I + K)^{-1} = (I + K)^{-1} \{ (I + K) - I \} (I + K)^{-1} \quad (13.7)$$

$$= (I + K)^{-1} K (I + K)^{-1} \succeq 0, \quad (13.8)$$

assuming $I + K \succ 0$ and $K \succeq 0$. For a one-dimensional quadratic example with information $a > 0$ and penalty $k \geq 0$, inverse curvature is $1/(a + k)$, whereas the sampling variance is $a/(a + k)^2$ and the local bias is $-k\theta_0/(a + k)$. Neither variance includes the squared bias needed for mean squared error about θ_0 .

13.3 Selected smoothing parameters, null spaces, and numerical ridge

All previous penalty derivatives hold λ fixed. If λ is selected from the same data, uncertainty in $\hat{\lambda}$ and its dependence on the regression estimate are omitted by a fixed- λ calculation. For a smooth solution branch, differentiating $U(\hat{\theta}) - K(\lambda)\hat{\theta} = 0$ gives

$$(H + K) \frac{\partial \hat{\theta}}{\partial \lambda_a} = -\frac{\partial K}{\partial \lambda_a} \hat{\theta}.$$

A joint delta calculation must include the variability of $\hat{\lambda}$ and the cross covariances with the fixed-smoothing fit, not merely append an independent variance term. A bootstrap that repeats the selection rule is another possible route under its own assumptions.

When $H \succeq 0$ and $K \succeq 0$,

$$v^\top (H + K)v = 0 \iff v \in \ker(H) \cap \ker(K).$$

The implication uses the nonnegativity of both terms and the spectral characterization of the null space of a PSD matrix. Thus $H + K \succ 0$ precisely when their null spaces have no common nonzero direction. A roughness penalty that leaves constant maps unpenalized cannot resolve a likelihood confounding direction that is also constant. For an indefinite H , this PSD null-space argument does not apply.

For example, if $Bc_B = \mathbf{1}_N$ and $z_i^\top c_\gamma = 1$ for every study, then shifting every group map by tc_B and shifting γ by $-tc_\gamma$ leaves the predictor unchanged. Likelihood identification requires a constraint or removal of the redundant direction. Penalizing that direction can select a unique answer, but it introduces additional assumptions rather than new data information.

Adding a numerical ridge εI_{n_θ} solely during covariance calculation changes the inverse being reported without changing the estimating equation that produced the fit. It is not automatically a fitted penalty or a sampling covariance for that fit. The repository's `ridge` option is explicitly numerical regularization. Its magnitude and inferential effect should be reported rather than described as magical identification.

14 Joint nuisance dispersion and reparameterization

14.1 Regression uncertainty after estimating dispersion

Now let $\alpha \in (0, \infty)^L$ collect dispersion parameters; $L = 1$ is allowed. Use a distinct joint information matrix

$$\mathcal{H} = \begin{pmatrix} H_{\theta\theta} & H_{\theta\alpha} \\ H_{\alpha\theta} & H_{\alpha\alpha} \end{pmatrix}.$$

Here $H_{\theta\theta}$ is the regression curvature with dispersion held fixed at its evaluation value, not the inverse-covariance block of a joint fit. If the nuisance block is nonsingular, eliminate the nuisance variables to define

$$H_{\text{eff}} = H_{\theta\theta} - H_{\theta\alpha} H_{\alpha\alpha}^{-1} H_{\alpha\theta}, \quad (\mathcal{H}^{-1})_{\theta\theta} = H_{\text{eff}}^{-1}. \quad (14.1)$$

This is the same Schur identity with the roles of the block variables reordered. If $\mathcal{H} \succ 0$, the subtraction is PSD and $H_{\text{eff}}^{-1} \succeq H_{\theta\theta}^{-1}$. For one nuisance parameter with cross column h and scalar curvature $a > 0$, the effective information is $H_{\theta\theta} - hh^\top/a$. A small nuisance dimension does not guarantee a small adjustment.

Expected information may have zero regression–dispersion cross blocks for some parametrizations and models. That must be established for the particular likelihood; it cannot be inferred from alternating optimization, and it does not force the observed cross block to be zero. For robust inference, stack the regression and dispersion estimating equations, form their joint bread and joint score meat, and then project the full sandwich onto regression coordinates. Replacing a regression bread by a Schur complement alone while ignoring nuisance-score variation is not a general robust adjustment.

The regression information and covariance interfaces discussed here in `cbmr_improved` hold the fitted dispersion fixed. They do not supply the joint nuisance adjustment in Equation (14.1). A likelihood fit that updates dispersion is not, by that fact alone, accompanied by joint-dispersion standard errors.

14.2 Log dispersion and the extra Hessian term

For positive dispersions write $\delta_a = \log \alpha_a$, so $\alpha_a = e^{\delta_a}$, and put $A_\alpha = \text{diag}(\alpha)$. Let $U_\alpha = \nabla_\alpha \ell$ at the evaluation point. The first derivative transforms as $U_\delta = A_\alpha U_\alpha$. Differentiating its a th component with respect to δ_b gives

$$\frac{\partial(U_\delta)_a}{\partial \delta_b} = \mathbf{1}\{a = b\} \alpha_a (U_\alpha)_a + \alpha_a \alpha_b \frac{\partial^2 \ell}{\partial \alpha_a \partial \alpha_b}.$$

Changing sign to information therefore yields

$$H_{\delta\delta} = A_\alpha H_{\alpha\alpha} A_\alpha - \text{diag}(\alpha_a (U_\alpha)_a : a = 1, \dots, L), \quad H_{\theta\delta} = H_{\theta\alpha} A_\alpha. \quad (14.2)$$

For a scalar this reads $H_{\delta\delta} = \alpha^2 H_{\alpha\alpha} - \alpha U_\alpha$. The minus sign of the extra score term follows from the convention $H = -\nabla^2 \ell$.

At an interior joint stationary point $U_\alpha = 0$, the extra term vanishes and, with $J_{\text{par}} = \text{blkdiag}(I_{n_\theta}, A_\alpha)$, the joint observed information transforms by $\mathcal{H}_{(\theta, \delta)} = J_{\text{par}}^\top \mathcal{H}_{(\theta, \alpha)} J_{\text{par}}$. Expected information obeys the corresponding rule under zero expected score and regularity. Away from stationarity, simply applying this congruence to an observed Hessian drops a real chain-rule term. At a penalized optimum the likelihood score alone may be nonzero: transform the full objective and its gradient if that is the curvature being used. Finally, $\alpha = 0$ is a boundary not represented by finite δ ; interior Wald arguments do not automatically cover it.

15 Voxel contrasts, nonlinear maps, and inferential calibration

15.1 SV coefficient ordering and tensor-product contrasts

Let $C_Z \in \mathbb{R}^{k \times R}$ specify contrasts among SV maps. At voxel j , stacking the R log effects gives

$$\begin{pmatrix} B_j^\top \beta_1 \\ \vdots \\ B_j^\top \beta_R \end{pmatrix} = (I_R \otimes B_j^\top) \beta.$$

Multiplication by C_Z and the Kronecker mixed-product identity give

$$\widehat{\delta}_j = (C_Z \otimes B_j^\top) \widehat{\beta} - \delta_{0j}, \quad V_{\delta j} = (C_Z \otimes B_j^\top) V_{\beta\beta} (C_Z^\top \otimes B_j). \quad (15.1)$$

The contrast matrix is $k \times RP$, not $RP \times k$. If global coefficients are present, append $k \times Q$ zeros to obtain a contrast on θ . For example, a row with entries $(1, -1, 0, \dots, 0)$ produces the coefficient row $[B_j^\top, -B_j^\top, 0, \dots, 0]$. This explicitly checks that the stacking convention agrees with the tensor-product order. Contrast row sums need not equal one; differences usually sum to zero.

15.2 GC group differences and global tests

With $C_G \in \mathbb{R}^{k \times G}$, use $\mathcal{T}_{Gj} = [C_G \otimes B_j^\top \quad 0_{k \times Q}]$. Then $\widehat{\delta}_j = \mathcal{T}_{Gj} \widehat{\theta} - \delta_{0j}$ and $V_{\delta j} = \mathcal{T}_{Gj} V \mathcal{T}_{Gj}^\top$, where V must be the covariance choice appropriate to the estimator. For one difference $\xi_g - \xi_h$, expansion gives

$$\text{Var}\{B_j^\top (\widehat{\xi}_g - \widehat{\xi}_h)\} = B_j^\top (V_{gg} + V_{hh} - V_{gh} - V_{hg}) B_j \quad (15.2)$$

$$= B_j^\top (D_g^{-1} + D_h^{-1}) B_j \quad (15.3)$$

$$+ B_j^\top (T_g - T_h) S^{-1} (T_g - T_h)^\top B_j, \quad (15.4)$$

where the last two lines assume the observed inverse-curvature choice and $g \neq h$. This expression retains shared-global uncertainty and shows when it cancels from a difference. Unequal study numbers per group affect the information but do not invalidate the identity.

A global contrast $C_\gamma \gamma$ has local model-based covariance $C_\gamma S^{-1} C_\gamma^\top$, not $C_\gamma \mathcal{E}^{-1} C_\gamma^\top$. A comparison of fitted intensities at different global covariate values must include the appropriate nonzero global contrast as well; it is not necessarily a pure group-map contrast.

15.3 Exponentiation and normalized intensities by the delta method

For one map let $\eta = B\xi$ and $\mu = \exp(\eta)$ componentwise. Since $\partial \mu_j / \partial \xi^\top = \mu_j B_j^\top$, the exact Jacobian is $J_\mu = \text{diag}(\mu)B$. The first-order expansion

$$\mu(\hat{\xi}) - \mu(\xi_0) \approx J_\mu(\xi_0)(\hat{\xi} - \xi_0)$$

leads to $\widehat{\text{Cov}}\{\mu(\hat{\xi})\} \approx J_\mu(\hat{\xi}) V_{\xi\xi} J_\mu(\hat{\xi})^\top$. The Jacobian is exact, whereas the covariance approximation is not. For a scalar log contrast δ , the ratio $r = e^\delta$ has approximate variance $r^2 \text{Var}(\hat{\delta})$. Exponentiating a log-scale interval often respects positivity better than a symmetric interval on the ratio scale.

For normalized spatial intensity define $p_j = \mu_j / m_\mu$, where $m_\mu = \sum_l \mu_l$. The quotient rule gives the entire normalization derivative:

$$dp_j = \frac{m_\mu d\mu_j - \mu_j \sum_l d\mu_l}{m_\mu^2} \quad (15.5)$$

$$= p_j \left(d\eta_j - \sum_l p_l d\eta_l \right), \quad (15.6)$$

$$J_p = \{\text{diag}(p) - pp^\top\}B. \quad (15.7)$$

Thus $\text{Cov}\{p(\hat{\xi})\} \approx J_p V_{\xi\xi} J_p^\top$. Because $\mathbf{1}_N^\top J_p = 0$, this covariance is singular in the sum-to-one direction and has rank at most $\min(P, N - 1)$. A common multiplicative global intensity factor cancels from this within-map normalization exactly. This does not mean global estimation has no indirect effect on the covariance of the fitted spatial coefficients. Covariances between normalized group maps still require $J_{p,g} V_{gh} J_{p,h}^\top$.

15.4 Wald tests: algebra versus the limiting null law

For a k -dimensional contrast with nonsingular estimated covariance, the Wald statistic is computed by solving $V_{\delta j} a = \hat{\delta}_j$ and taking

$$W_j = \hat{\delta}_j^\top a = \hat{\delta}_j^\top V_{\delta j}^{-1} \hat{\delta}_j.$$

If the null is true, the estimator is asymptotically normal in the tested directions, the covariance estimate is consistent, and the k constraints are independent, then W_j converges in distribution to χ_k^2 . For one contrast the signed statistic is $Z_j = \hat{\delta}_j / \sqrt{V_{\delta j}}$, with $W_j = Z_j^2$. These distributional statements are not exact consequences of Schur inversion.

Redundant constraints should be reduced to an independent set. A generalized inverse can describe a known lower-dimensional estimable test under additional rank conditions, but cannot rescue unidentified contrasts, too few independent clusters, or an unstable covariance. Smoothing bias, estimated nuisance parameters, boundary fits, and data-dependent model selection all require attention. Pointwise voxel p -values are not automatically multiplicity-adjusted or simultaneous spatial inference. Faster exact linear algebra does not establish a universal minimum number of foci or studies for Wald calibration.

15.5 Bootstrap units, null fitting, and repeated fitting decisions

A bootstrap must reproduce the estimator and the relevant sampling mechanism. For a nonparametric study bootstrap, resample independent studies (or larger independent clusters), retaining each selected unit's entire voxel pattern and covariates. Group-stratified resampling may be appropriate when the design conditions on group counts. Resampling individual voxels instead changes the within-study dependence model. A fixed-design parametric bootstrap instead keeps covariates and group membership fixed and simulates new responses.

For a fitted-null parametric test, the operational sequence is: fit the stated null with its constraints; simulate under that null and its dependence model; refit each simulated dataset by the original procedure; and recompute the same statistic. If the procedure estimates dispersion or selects smoothing parameters, repeat those steps unless the explicit target is conditional on their fixed values. A shared study latent effect must be shared across that study's simulated voxels. Independent-cell NB and aggregate-NB sampling schemes are not interchangeable with a study-clustered NB scheme.

With B_{boot} successful replicates, a common nonzero Monte Carlo tail estimate is

$$\hat{p} = \frac{1 + \sum_{b=1}^{B_{\text{boot}}} \mathbf{1}\{W_b^* \geq W_{\text{obs}}\}}{B_{\text{boot}} + 1}.$$

The statistic and tail must match the alternative, and failed refits must be diagnosed rather than silently discarded as if random. The +1 convention does not make a fitted-null parametric bootstrap an exact finite-sample test. Bootstrap validity and any tail extrapolation remain separate statistical questions.

16 Conditioning, Kronecker structure, and computational accounting

16.1 Deriving the spectral condition number

For a real symmetric nonsingular H , take the orthogonal eigendecomposition $H = O\Lambda O^\top$, with $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_n)$. Then

$$H^\top H = O\Lambda^2 O^\top.$$

The singular values are the square roots of the eigenvalues of this product, namely $|\lambda_a|$. Therefore

$$\|H\|_2 = \max_a |\lambda_a|, \quad \|H^{-1}\|_2 = \frac{1}{\min_a |\lambda_a|}, \quad \kappa_2(H) = \frac{\max_a |\lambda_a|}{\min_a |\lambda_a|}. \quad (16.1)$$

For SPD H , absolute values can be dropped. For an indefinite matrix, using the largest signed eigenvalue divided by the smallest signed eigenvalue is wrong. A zero eigenvalue makes the matrix singular and the ordinary condition number infinite. The example $\text{diag}(1, -1)$ has $\kappa_2 = 1$ but cannot be a covariance precision for a nondegenerate Gaussian distribution.

For a block diagonal matrix, $\det(tI - D) = \prod_g \det(tI - D_g)$, so its spectrum is the union of block spectra and

$$\kappa_2(D) = \frac{\max_g \max_a |\lambda_a(D_g)|}{\min_g \min_a |\lambda_a(D_g)|}.$$

This is not generally $\max_g \kappa_2(D_g)$. For example, $D_1 = I_P$ and $D_2 = 10^6 I_P$ each have condition number one, whereas $\text{blkdiag}(D_1, D_2)$ has condition number 10^6 . The scale across blocks matters even when each block is internally well conditioned.

For a bordered system, Equation (10.16) preserves inertia, not eigenvalues. In the scalar-block example $D = 1$, $F = t$, $\mathcal{E} = 1 + t^2$, the Schur complement is $S = 1$. Both factors have condition

number one, but the full matrix has determinant one and trace $t^2 + 2$. Its eigenvalues are

$$\lambda_{\pm} = \frac{t^2 + 2 \pm \sqrt{(t^2 + 2)^2 - 4}}{2},$$

so $\kappa_2(H) = \lambda_+/\lambda_-$ grows without bound as $|t|$ increases. A full symmetric eigenproblem is required for the exact bordered spectral condition number.

Conditioning depends on coefficient units. Under a nonsingular linear reparameterization $\theta = L\phi$, the information becomes $H_{\phi} = L^{\top} H_{\theta} L$ and can have a different condition number, while correctly transformed contrasts describe the same estimand. For fixed H , perturbing a right-hand side gives the bound $\|\Delta x\|_2/\|x\|_2 \leq \kappa_2(H)\|\Delta b\|_2/\|b\|_2$. This follows from $\Delta x = H^{-1}\Delta b$ and $\|b\|_2 \leq \|H\|_2\|x\|_2$; it explains numerical sensitivity, not statistical validity. A one-norm estimator such as LAPACK's `pocon` estimates a different norm condition number and is not an exact computation of κ_2 .

16.2 When a Kronecker inverse is justified

For the SV independent-cell information without a global border, Equation (9.4) can be written

$$H = \sum_{i,j} w_{ij} (Z_i Z_i^{\top}) \otimes (B_j B_j^{\top}).$$

If the weights really separate as $w_{ij} = a_i b_j$, distributivity of the tensor product gives the exact factorization

$$H = A_Z \otimes A_B, \quad A_Z = \sum_i a_i Z_i Z_i^{\top}, \quad A_B = \sum_j b_j B_j B_j^{\top}. \quad (16.2)$$

This is a sufficient separability condition, not a consequence of having written an unweighted design as a Kronecker product. Generic SV means vary with both study and voxel through several interacting maps. Independent-NB denominators and observed curvature weights can destroy a mean factorization as well. A sum of Kronecker products is not generally a single Kronecker product. A global border or a penalty also requires an explicit new structural check.

If the two square factors are nonsingular, the mixed-product rule verifies the inverse:

$$(A_Z \otimes A_B)(A_Z^{-1} \otimes A_B^{-1}) = (A_Z A_Z^{-1}) \otimes (A_B A_B^{-1}) = I_R \otimes I_P.$$

The reverse product is the identity too. If either factor is singular, a nonzero null vector in that factor produces a null vector of the product, so an ordinary inverse is unavailable. Singular values of a Kronecker product are pairwise products of singular values of its factors; consequently, for nonsingular factors,

$$\kappa_2(A_Z \otimes A_B) = \kappa_2(A_Z) \kappa_2(A_B).$$

For symmetric factors this also follows from tensor products of their orthonormal eigenvectors and products of absolute eigenvalues.

To solve without forming the product, reshape a right-hand side as $b = \text{vec}(Y)$ with $Y \in \mathbb{R}^{P \times R}$, stacking columns in the same map-major order as β . The identity

$$(A_Z \otimes A_B) \text{vec}(X) = \text{vec}(A_B X A_Z^{\top})$$

turns the equation into $A_B X A_Z^{\top} = Y$. First solve $A_B W = Y$, then solve $A_Z X^{\top} = W^{\top}$. Using an approximate separable matrix as a preconditioner can be useful, but its inverse is not the exact covariance of the original information.

16.3 Operation counts, storage, and implementation scope

The following counts concern linear algebra *after* information assembly, use ordinary dense block arithmetic, and omit constant factors. Let block g have size d_g , with $\sum_g d_g = d$; use component indices in place of group indices for general SV designs.

Representation	Initial factorization work	Factor storage
Full dense, $n_\theta = d + Q$	$O(n_\theta^3)$	$O(n_\theta^2)$
Unbordered blocks	$O(\sum_g d_g^3)$	$O(\sum_g d_g^2)$
Bordered blocks	$O(\sum_g d_g^3 + Q \sum_g d_g^2 + dQ^2 + Q^3)$	$O(\sum_g d_g^2 + dQ + Q^2)$
Exact $A_Z \otimes A_B$	$O(R^3 + P^3)$	$O(R^2 + P^2)$

For the bordered row, the terms arise in order: factor every D_g ; solve $D_g T_g = F_g$ for Q right-hand sides; form $S = \mathcal{E} - F^\top T$; and factor S . Afterwards, solving k general right-hand sides costs

$$O\left(k \sum_g d_g^2 + kdQ + kQ^2\right).$$

The block triangular solves account for the first term, $F^\top y$ and Tx_γ for the second, and the Schur solves for the third. A full dense solve after factorization costs $O(kn_\theta^2)$. For the Kronecker system, the two factor solves cost $O(P^2 R + PR^2)$ per right-hand side. Holding its reshaped right-hand side requires $O(PR)$ additional storage.

Selected covariance output has its own costs. Storing X in Equation (11.1) requires $O(n_\theta k)$, and a dense multiplication CX costs $O(n_\theta k^2)$. If the full spatial correction is demanded, solving $SW = T^\top$ costs $O(Q^2 d)$ and forming TW costs $O(d^2 Q)$; storing the resulting full covariance requires $O(n_\theta^2)$. Computing all diagonal entries avoids that output allocation, though generic block-inverse diagonals may still need $O(\sum_g d_g^3)$ work via solves against block identity matrices.

For projected study sandwiches, forming UX costs $O(Mn_\theta k)$ with dense score rows and then its Gram matrix costs $O(Mk^2)$. If scores arrive study by study, one can accumulate $v_i v_i^\top$ immediately and retain only $O(k^2)$ output storage, in addition to the factors and the contrast solves. Model-specific sparse projections can reduce score-assembly work. These are order calculations, not empirical performance claims; likelihood assembly, input storage, sparsity, and numerical libraries affect actual timings.

The SPD path in `cbmr_improved/structured.py` caches block Cholesky factors, computes T by solves, and factorizes the Schur complement. Its contrast, diagonal, and score-Gram operations implement the algebra above without a full spatial inverse on that path. If block or Schur Cholesky fails, it logs a fallback to a dense symmetric solve; an invertible indefinite matrix can then be solved algebraically, while a singular full system raises an error. The structured savings need not survive this fallback. Its full covariance compatibility method deliberately solves against the full identity.

The same implementation computes the spectral condition number from block eigenvalues when there is no global border, and from a full dense symmetric eigenproblem when there is one. That diagnostic itself can require $O(n_\theta^2)$ memory and $O(n_\theta^3)$ work even if the intended contrast solve is structured. Exact algebraic identities do not remove floating-point error, establish statistical calibration, or constitute formal certification of this explanatory fragment.

16.4 Manual audit prompts

1. Which regression coefficients and nuisance parameters were actually estimated? Does the stated covariance condition on any of them?

2. Is the matrix observed information, expected information, penalized curvature, or an estimating-equation sensitivity? Are its sign and summed-versus-averaged normalization explicit?
3. Can every claimed structural zero be traced to factor support or a specified penalty? Are entire connected components, not isolated missing edges, used to choose blocks?
4. Do the products $DT = F$, $S = \mathcal{E} - F^\top T$, and $HX = C^\top$ have small residuals at the numerical fit? Are the matrices whose positive definiteness is required actually SPD?
5. Are group cross covariances retained, and does a selected contrast use the same coefficient ordering as $C_Z \otimes B_j^\top$ or $C_G \otimes B_j^\top$?
6. For a sandwich, are the score rows derivatives of the chosen likelihood or estimating equation, grouped by genuinely independent units? Is their rank sufficient for the proposed test?
7. For a penalty or numerical ridge, what changes in the estimator, bias, covariance interpretation, and identification assumptions? Was smoothing selection uncertainty intentionally included or omitted?
8. Does a proposed Kronecker factorization hold for the actual weights at this fit, rather than only for the unweighted design?
9. Which statements are exact matrix identities, which are first-order approximations, and which rely on a bootstrap or empirical calibration? Have finite-sample validity and multiplicity been assessed separately from computational speed?

17 Source correspondence and manual review checklist

17.1 How this document relates to the repository

Material	Related source
Poisson scalar and structured Hessians	<code>proofs/poisson.py</code> , <code>proofs/appendix.py</code>
Independent NB scores, observed/Fisher weights, SV blocks	<code>proofs/appendix.py</code>
Aggregated NB (ρ, A) partials and inner chain rule	<code>proofs/negative_binomial.py</code>
Clustered NB and grouped likelihood reductions	<code>proofs/clustered_negative_binomial.py</code> , <code>proofs/published_cbmr.py</code>
Schur, block and contrast identities	<code>proofs/covariance.py</code> , <code>proofs/appendix.py</code> , <code>proofs/published_cbmr.py</code>
Structured solves and stable aggregate derivatives	<code>cbmr_improved/structured.py</code> , <code>cbmr_improved/information.py</code>
Existing formal proof coverage and its limitations	<code>docs/lean_coverage.md</code>

The notation ρ in this document corresponds to shape R in the original aggregated-NB proof code; it is not the number R of SV maps. Original implementation names such as `observed` `Fisher` `information` refer here to observed information H , not automatically expected Fisher information I .

The correspondence table is a reading aid, not an assertion that every extension here is covered by those proofs. In particular, the expanded nuisance calculations, general SV moment-matching composition, and handwritten intermediate arguments are not automatically certified by the existing integrated-appendix Lean inventory. No empirical timing or calibration result is inferred from a matrix identity.

17.2 A suggested order for a line-by-line audit

First fix the chosen statistical model and the ordering of θ ; write down one design row x_{ij} and confirm its dimension. Check whether a covariate has a scalar coefficient or a spatial map. Next verify the scalar score and negative second derivative before looking at any matrix formula. For independent NB check the quotient rule numerator; for aggregated NB check *both* inner Hessians; for clustered NB check the derivative of the total t_i .

For each information formula, confirm whether it is observed or expected and which dispersion parameter was held fixed. For each data reduction, identify precisely which counts remain. If a reduction discards study totals, compare it with (3.12). If it uses an NB total, determine whether its distribution is exact or moment matched.

Before reading a covariance as an inferential answer, check identifiability, nuisance adjustment, the choice of bread and meat, and whether a penalty was used. For group contrasts retain covariance induced by shared global effects. For a claimed Kronecker inverse check weight separability, not just the use of a tensor-product basis. For any Wald statement distinguish pointwise inference from simultaneous control over many voxels.

17.3 Rebuilding the separate PDF

From the repository root, run

```
make expanded-appendix
```

The editable main source is `docs/cbmr_expanded_appendix.tex`; its covariance chapters are in `docs/expanded_covariance.tex`. The build writes `docs/cbmr_expanded_appendix.pdf` without rebuilding or replacing the existing appendix or reference PDFs. Three LaTeX passes populate the contents and settle its page cross-references.